COMPETITION & DESIGN PROCESS GUIDELINES

30.01.2026 VERSION 1.0

INTENTION

Structure key information from **brief, design planning and parameters**

Ensure **key information** is **visible and accessible** throughout the competition

Support **early understanding** of the key factors influencing concept exploration

LAVA PROCESS

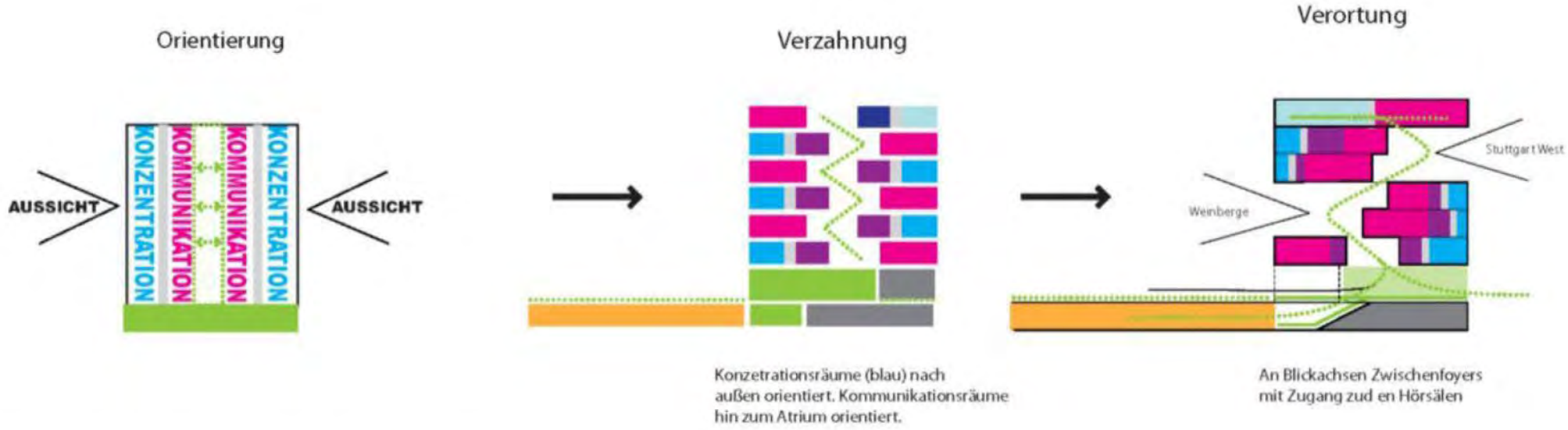
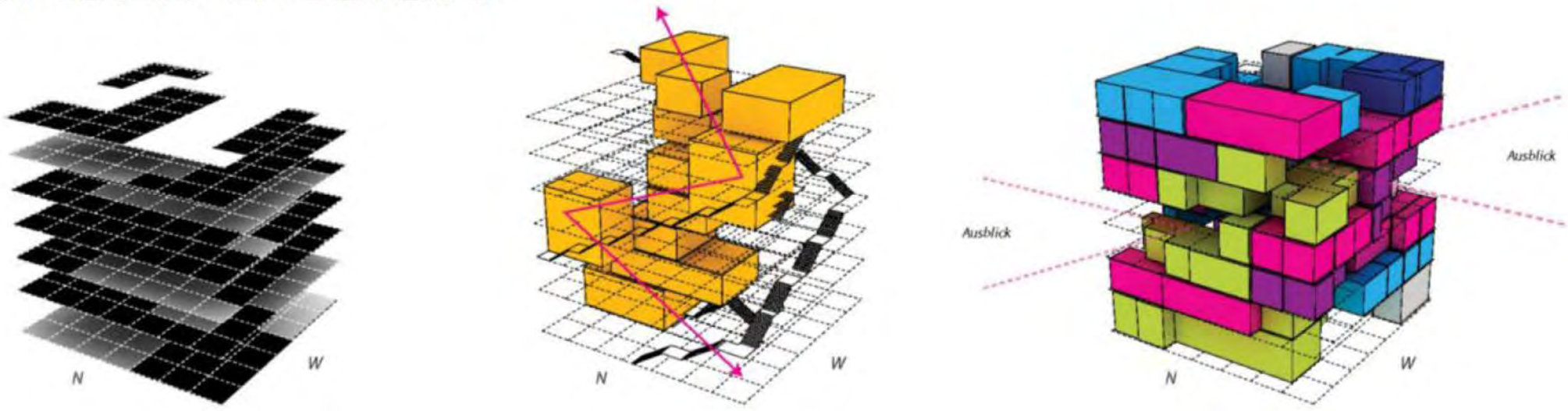


DIAGRAM BASED DESIGN



Dichte und offene Zonen

Luftraum

Programmverteilung


What if architecture,
informed by scientific inquiry,
could spark our imagination
and redefine our understanding
of the built environment?
Bottom up Analysis
Top Down Idea
Conceptual Innovation
L/\V/\
LABORATORY FOR
VISIONARY
ARCHITECTURE

Inquire. Imagine. Invent.

:to investigate; look into

:to form a mental image of (something not present)

:to produce (something, such as a useful device or process) for the first time through the use of the imagination or of ingenious thinking and experiment

L/\\//\

PROJECT SETUP & DESIGN PROCESS

Admin

Application requirements

- Application forms
- Office information
- Reference projects
- Contact consultants
- CVs preparation

Submission requirements

- Digital Submission
- File format, size
- Naming conventions
- Print out and post service

Design & Delivery

Design Planning

- Establish weekly goal
- Weekly meeting
- Minutes of meeting
- Deliverable lists

Understanding the Vision

- Design Brief understanding
- Set up internal project scope (Research, test of processes)
- Benchmark analysis

Understanding the team

- Past projects from office
- Experts/ knowledge sharing
- Assign tasks and responsibilities

CHECKLISTS

Ensure **critical tasks, documents, and decisions** are not missed
under competition time pressure

LAVA

GENERAL PLANNING - CHECKLIST

Checklist - Adjust based on requirements and competition durations

00_BRIEF CHECKLIST

Dates & Milestones

01_APPLICATION CHECKLIST

Application and relevant documents, Eligibility

02_SUBMISSION CHECKLIST

Submission format and list

03_DEVELOMENT PARAMETERS

Planning, program, technical requirements

04_MINUTES OF MEETINGS

Records, decisions

GENERAL PLANNING - CHECKLIST

Checklist - Adjust based on requirements and competition durations

00_BRIEF CHECKLIST

LAVA

HIGH-LEVEL PLANNING CHECKLIST

Project Number:	xxx	Project Title:	xxx
Date:	27/01/26	Stage:	Competition

BRIEF	Description	Dates
	IMPORTANT INFO	
Application deadline		YYMMDD
Submission deadline		YYMMDD
Q&A Deadline		YYMMDD
Site Visit (if any)		YYMMDD
Interim presentation (if any)		YYMMDD
Final presentation (if any)		YYMMDD
Result annoucement		YYMMDD
Anonymity requirements	Yes/No	
Local architect required	Yes/No	
Language of submission	English/ Bilingual	

GENERAL PLANNING - CHECKLIST

Checklist - Adjust based on requirements and competition durations

01_APPLICATION CHECKLIST

LAVA

HIGH-LEVEL PLANNING CHECKLIST

Project Number: xxx Project Title:
Date: 27/01/26 Stage:

ADMIN	Description	Deadline	Status
	APPLICATION REQUIREMENTS		
Application forms			
-Form 1			
-Form 2			
-Form 3			
Prepare Office CV	Only if required by the brief		
-Office descriptions 1			
-Office descriptions 2			
Compile office reference project	Only if required by the brief		
- Key projects 1			
- Key projects 2			
- Key projects 3			
Contact consultants	Only if required		
- Structural Engineer (XYZ)			
- MEP Consultant (XYZ)			
- Landscape architect (XYZ)			
- AOR			
- EOR			
- CGI			
Team members CVs	Only if required by the brief		
- Member 1			
- Member 2			
- Member 3			

02_SUBMISSION CHECKLIST

LAVA

HIGH-LEVEL PLANNING CHECKLIST

Project Number: xxx Project Title:
Date: 27/01/26 Stage:

ADMIN	Description	Deadline	Status
	SUBMISSION REQUIREMENTS		
List out submission materials			
-Boards	6 panels (For example)		
-Video	3 minute video (For example)		
-Images	1920 x 2080 (For example)		
-Physical model	Model base with 2m x 2m (For example)		
-BIM model			
-CAD drawing			
-Project texts	10 pages (For example)		
-Cost estimation			
List out submission materials format	File size, Landscape/ Portrait		
-Boards	Landscape 1000x700mm (For example)		
-Video	-1080 x 720 (For example)		
-Physical model	1:500 model with context (For example)		
-BIM model			
-CAD drawing			
List out naming conventions	From brief requirements		
-Boards			
-Video			
-Physical model			
-BIM model			
-CAD drawing			
Contact output shop	Find out expected printing time		
-Output shop name 1	6 hours printing for 6 panels, with mounting on foam (for example)		
Contact post service	Find out expected posting time		
-Post name 1	1.5 days deliver to Casablanca (for example)		
Contact model maker			
-Model maker 1			

GENERAL PLANNING - CHECKLIST

Checklist - Adjust based on requirements and competition durations

03_DEV. PARAMETERS

LAVA

HIGH-LEVEL PLANNING CHECKLIST

Project Number: xxx Project Title:
Date: 27/01/26 Stage:

DEVELOPMENT PARAMETERS	Description		
	SITE		
Site area	xxx sqm		
Buildable area	xxx sqm		
Plot ratio / FAR	x		
Site coverage	x%		
Setbacks	x m		
Open space area	xxx sqm		
Landscape area	xxx sqm		
	MASSING		
GFA	xxx sqm		
BUA	xxx sqm		
Height limit	x m		
Basement depth	x m		
Podium height	x m		
	PROGRAM DISTRIBUTION		
Residential	xxx sqm		
Retail	xxx sqm		
Office	xxx sqm		
Gallery	xxx sqm		
Entertainment	xxx sqm		
Circulation	xxx sqm		
BoH	xxx sqm		
TOTAL AREA			
	TECHNICAL REQUIREMENT		
Floor to ceiling height	x m		
Floor plate depth	x m		
Exit distance	x m		
Number of elevators	x		
Number of escalators	x		
Number of stairs	x		
Number of occupancy	x		
Number of toilets	x		
	PARKING & MOBILITY		
Parking space			
Bicycle parking			
EV charging			
Loading bays			
	EXPECTED COST		
Total cost from brief			

04_MINUTES OF MEETING

LAVA

Minutes of Meeting

Meeting Purpose – Meeting No. 1

Project Title	Project Code
XXXX	xxx-xxxx
Date	**Location**
DD / MM / YYYY	LAVA Berlin, Saarbrücker Straße 24 Haus D

General Statement
Minutes of Meeting shall be deemed acceptable in the absence of written comments or corrections from participants within 14 days of distribution.

Meeting Agenda
Brief description of the objectives of the meeting, topics to be presented / issues to be resolved, etc.

Attendees	Acr.	Company	Role	Attendance (Y/N)
First Name Last Name		LAVA	Architect	Y

Nr	Topic	Action	Responsible	By (date)
1	Topic 1			
1.1	Sub-Topic 1 General notes and statement • Details if required • Details if required • Details if required	Takeaways from the meeting – feedback, actionable items and decisions	Consultant or Person Name	DD/MM/YYYY
1.2	Sub-Topic 2 General notes and statement • Details if required • Details if required • Details if required	Takeaways from the meeting – feedback, actionable items and decisions	Consultant or Person Name	DD/MM/YYYY

DESIGN TIMELINE SET UP

Adjust based on requirements and competition durations

PHASES	UNDERSTANDING OF BRIEF & REQUIREMENTS	CONCEPT EXPLORATION	CONCEPT DEVELOPMENT	PRODUCTION
TASKS	- Extraction of keywords from the brief - Reference projects Benchmark Strategy learning - Site Analysis - Research Culture, Environment, Traffic, City planning, History, Technical, Graphics	- Design options Strategy, typology, configuration - Initial massing tests - Initial plans, sections - Initial area check - Circulation principle - Structure principle - Building services principle	- Façade/floor/ wall/ ceiling options based on finalized massing - Materials options - Drawings development - Set up rendering angles	- Finalize board layout - Representations optimization - Project texts
OUTCOME	- Brief summary - Deliverable list - Agree on project vision/ Set up Benchmark - Typology understanding - Develop performance drivers - Consolidation of site understanding	- Decision on concept direction for further development - Initial board layout - Initial drawings set up	- Materials selections - Establish façade systems - Finalized render model - Estimation for production materials	- Submission materials ready to upload/ send
DURATION	- 1 week	- 2 – 3 weeks	- 1 – 2 weeks	- 1 – 2 weeks

Each phase conclude with a clear internal decision before proceeding

L/\V/\

DESIGN TIMELINE SET UP

Adjust based on requirements and competition durations

PROJECT SAMPLE : 499 BUR

LAVA

HIGH-LEVEL TIMELINE

Project Number:	499	**Project Title:**	BUR
Date:	2010/1/26	**Stage:**	Competition

499

	10.11	11.11	12.11	13.11	14.11	17.11	18.11	19.11	20.11	21.11	24.11	25.11	26.11	27.11	28.11	1.12	2.12	3.12	4.12	5.12	8.12	9.12	10.12	11.12	12.12	15.12	16.12	17.12	18.12	19.12
	week 1					week 2					week 3					week 4					week 5					week 6				
	mon	tue	wed	thu	fr	mon	tue	wed	thu	fr	mon	tue	wed	thu	fr	mon	tue	wed	thu	fr	mon	tue	wed	thu	fr	mon	tue	wed	thu	fr
Tasks/ Weekly Goal(s)	Concept Exploration					Concept Exploration Finalize					Concept Development					Concept Development					Concept Development, Façade and Detail					Drawing, Diagram, Panel			
Contextual reseach, density, circulation, vegetation																														
Program - Typology understanding																														
Benchmark analysis - Reference projects																														
Technical research, FLS requirements, climate																														
Local materials, culture and climatic response analysis																														
Develop key performance drivers																														
Site Model																														
Board Layout test																														
Concept diagram, key drivers, program and circulations																														
Reference project, mood images																														
Sketch Plan, organization																														
Massing test																														
Sketch Physical model																														
Internal Render test																														
Finalize massing																														
Initial plan drawings set up																														
Area checking																														
Façade strategy																														
Render angle set up																														
Explanation Diagram production																														
Texts, narrative																														
Drawings, plans, sections elevations																														
Translation and layout adjustment																														
Render, CGI comments																														
Internal Renders																														
Final Board layout																														

Weekly report & Minutes of meeting

Info Exchange: Design Direction

Weekly report & Minutes of meeting

Weekly report & Minutes of meeting

Info Exchange: Design Update

Weekly report & Minutes of meeting

CGI: Model send out

Weekly report & Minutes of meeting

Info Exchange: Translation, Production

3D GEOM

views

5% GEO

1ST DRAFT

comment

Send A&A - plans - Budget - tables

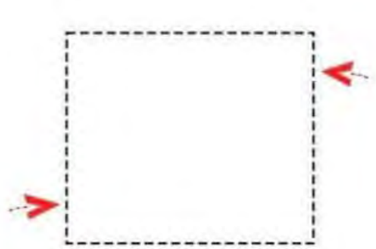
Landsca Send A&A email fo

Concept Exploration	2 week	
Concept Development	3 week	
Production & Presentation	1 week	

Each phase conclude with a clear internal decision before proceeding

PROJECT INITIATION

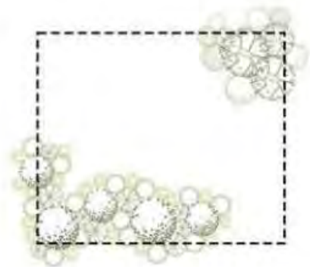
SITE ANALYSIS



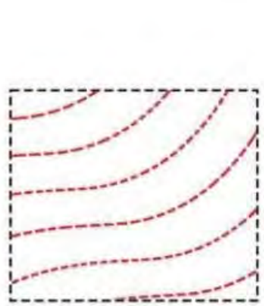
Points of access



Airflow



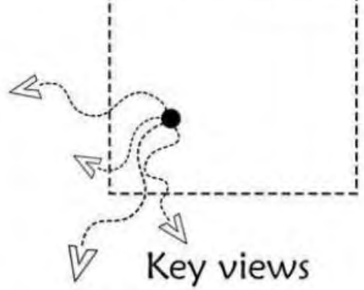
Native vegetation



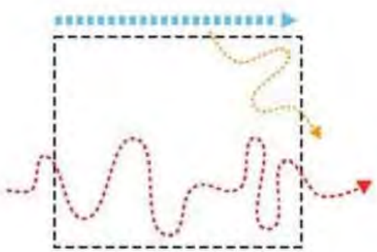
Topography



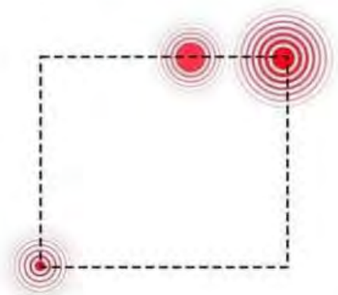
Sun direction



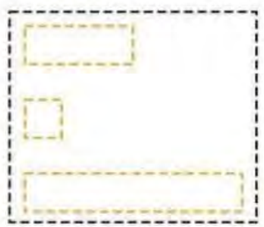
Key views



Circulation

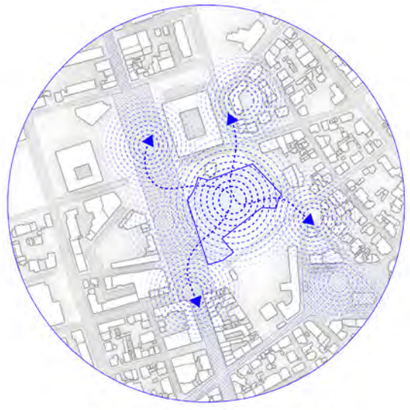


Noise nuisance



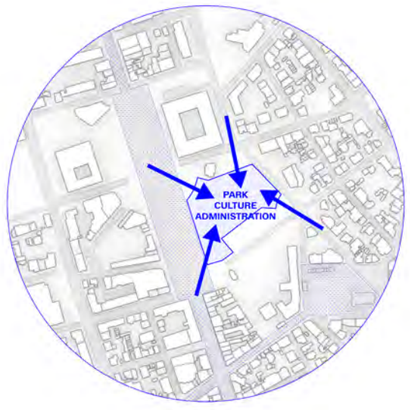
Existing framework

SITE SYNTHESIS



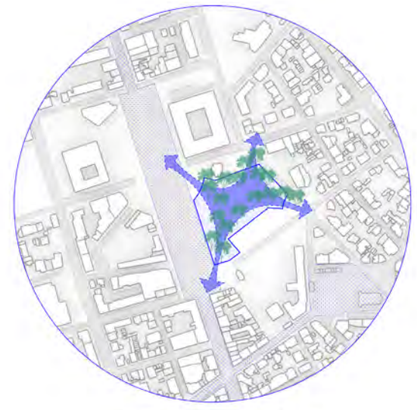
regeneration & activation

The municipality's goal is to **revitalize "The Hole"** and **the city center** by transporming a long-abandoned void to a **living, vibrant space.** The design is not just consturuction, but an urban healing. It must generate activity through flexible cultural programming, public interaction, and open access.



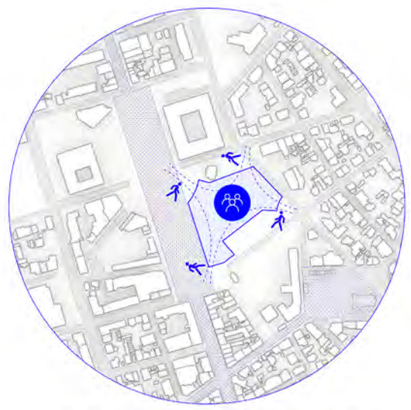
integration

The brief repeatedly calls for the **connection** of park, cultural, and administrative functions; not as isolated buildings but as a **coherent system**. The project must integrate with existing squares *(Troikata, Baba Ganka)*, the pedestrian network, and the city's urban fabric, forming one continuous public realm.



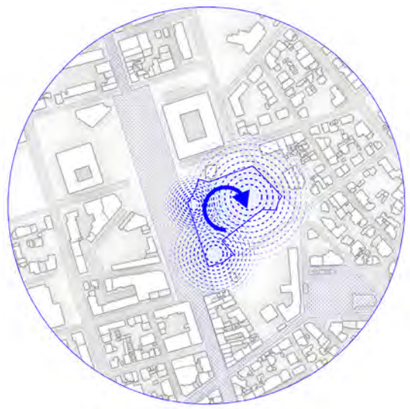
green infrastructure

The project must function as **a park environment**, improving microclimate, air quality, and comfort. It requires a **minimum 5,400 m² of greenery** (including 1,800 m² of trees) and should address urban heat island mitigation, water management, and overall environmental performance.



cultural and social ecosystem

The project is intended to become **a social connector**, a space that brings together diverse groups of people and strengthens community identity. The project must accommodate different scales and types of gathering, ensuring a balance between cultural, educational, and everyday social uses.Creating an inclusive civic landscape where **everyday life, culture, and learning overlap** encouraging encounters between residents, artists, and visitors, and strengthening the collective identity of Burgas.



coexistence & flexibility

The site's success depends on their mutual balance, where nature, everyday life, and cultural expression reinforce one another rather than compete. Spaces should allow for **different configurations and scales of use,** exhibitions, performances, education, workshops, debates. This includes both temporary and permanent functions, ensuring adaptability to the city's evolving cultural life.



exchange & accessibility

The brief and the public consultation both emphasize **dialogue, participation, and shared creation.**
This space should be a platform for exchange between artists and citizens, between the city and visitors. All spaces (open and closed) must ensure equal access for people with mobility, visual, and hearing impairments, while the site itself must strengthen pedestrian connections and ensure inclusive movement across the area.

PROJECT VISION

Our previous and future work is strategically framed through the lens of 8 key themes. When starting a project, it is important to determine the following :

- **Which LAVA Themes** relate the project and how ? Multiple themes can be applicable.
- **What is the envisioned lab approach ?** What are the tools and workflows we would like to explore and use ?
- **Which previous LAVA projects** can be studied and benchmarked ? Are there any precedents or approaches which we have learned from in the past ?

As the Project Team execute the work, Project Leadership controls that the above is integrated in the design.

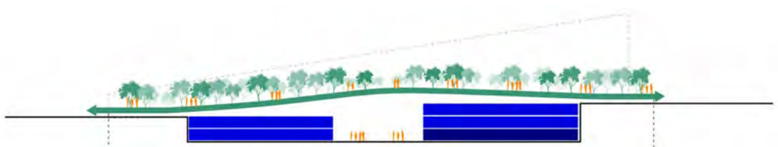
LAVA THEMES & STRATEGIES



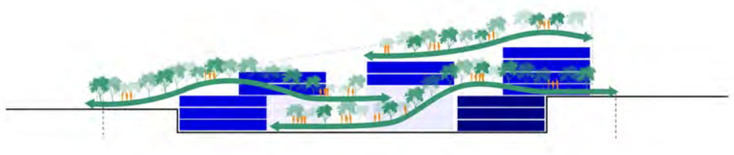
Architecture driven by technology, data, and AI revolutionizes both design and production.	Reimagining healthy and intelligent urban landscapes.	Private environments that supercharge health, happiness, comfort, and well-being through design.	Holistic, architectural ecosystems that replenish and enhance the earth beyond sustainability.	When explorative ideas animate architectural concepts, imaginations become agents of innovation.	When typologies and functions fuse, architecture forms multifaceted spaces beyond mix-use.	Creative energy thrives in connections and collaboration at the workplace of the future.	Public spaces and buildings designed for dynamic interactions and close-knit communities.
Digital Workflows	**Urban**	**Private**	**Sustainability**	**Experimental**	**Mixed-use**	**Workspace**	**Public Spaces**
• Technology • Data • AI • Digital Fabrication • etc.	• Healthy • Intelligent • Urban • Landscapes	• Private • Living • Health • Comfort • Well-being	• Ecosystems • Holistic • Sustainable • Earth	• Explorative • Conceptual • Imaginative	• Typologies & Functions fuse • Multifaceted • Multifunctional	• Collaborations • Workspace • Connectivity	• Public • Interactions • Dynamic • Communities

PROJECT BENCHMARKS & REFERENCE

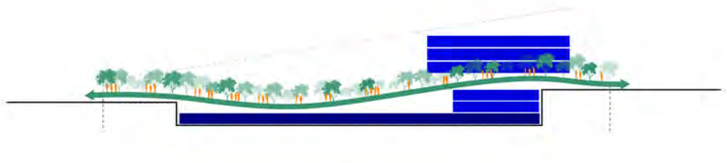
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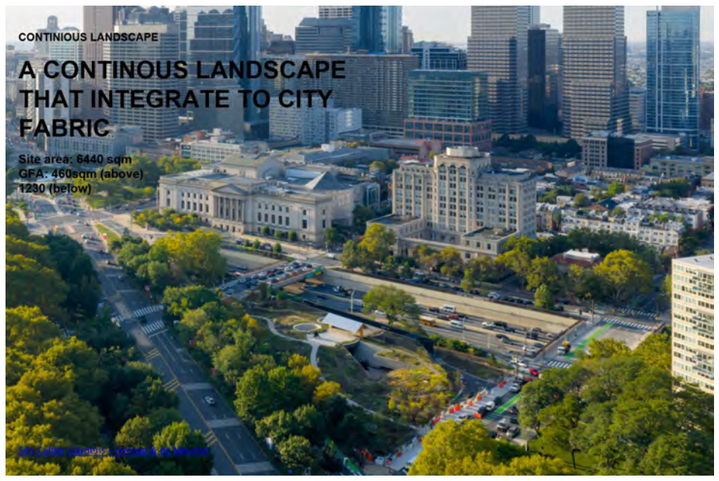


2



3









CONCEPT INITIATION

LAVA



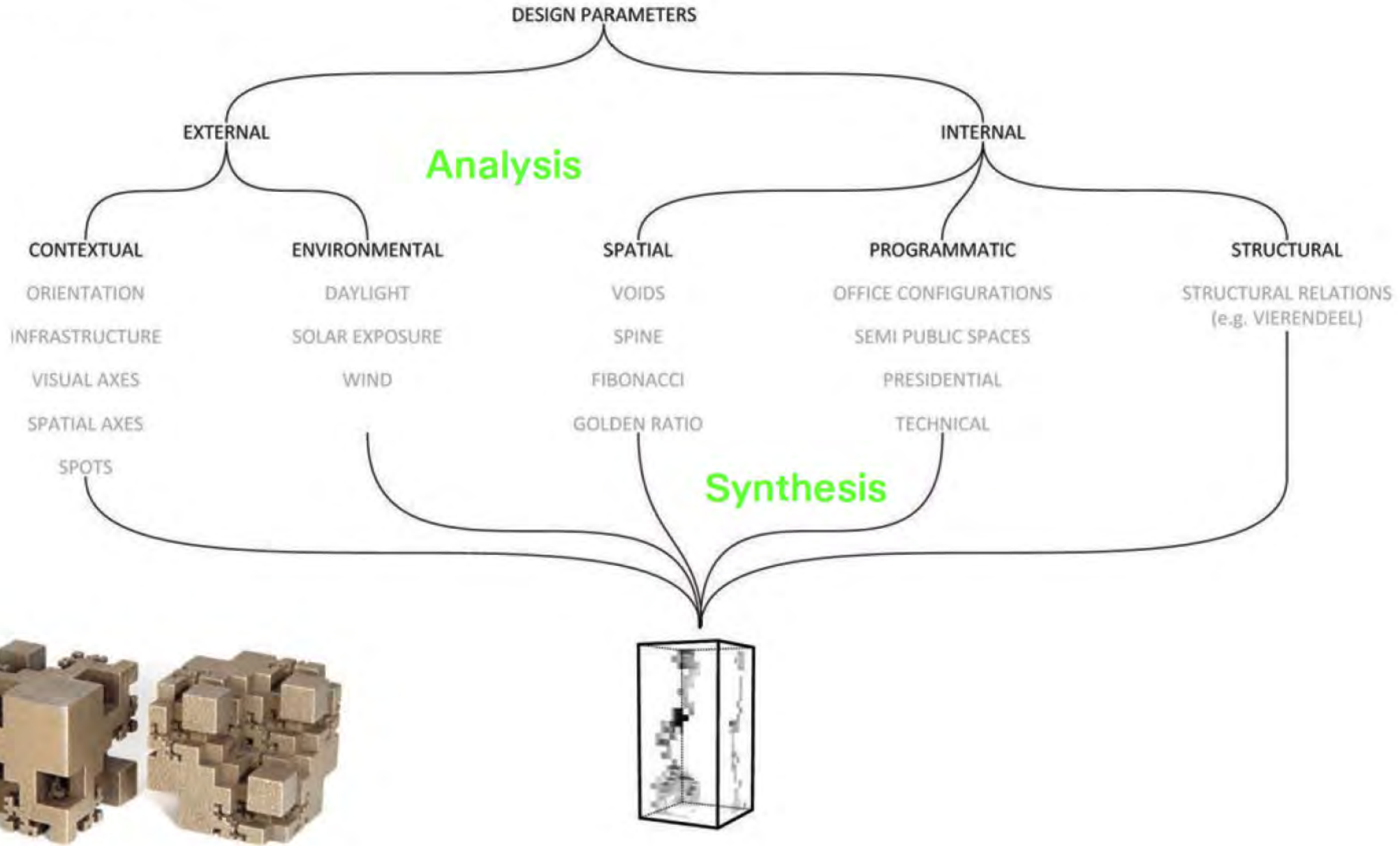
KEY QUESTION What if architectural form could facilitate communication?

LAVA



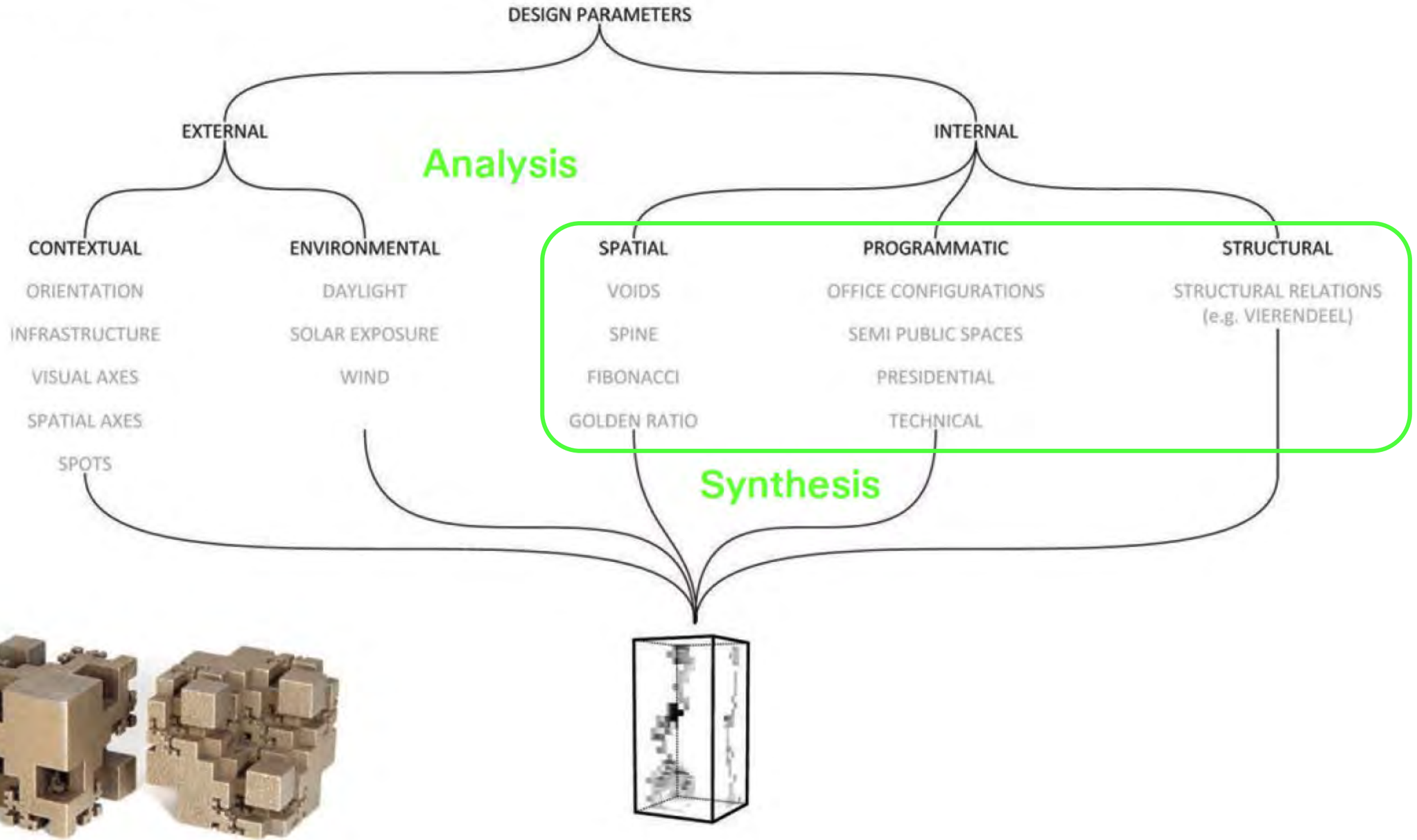
KACST INNOVATION TOWER

DESGIN PARAMETERS



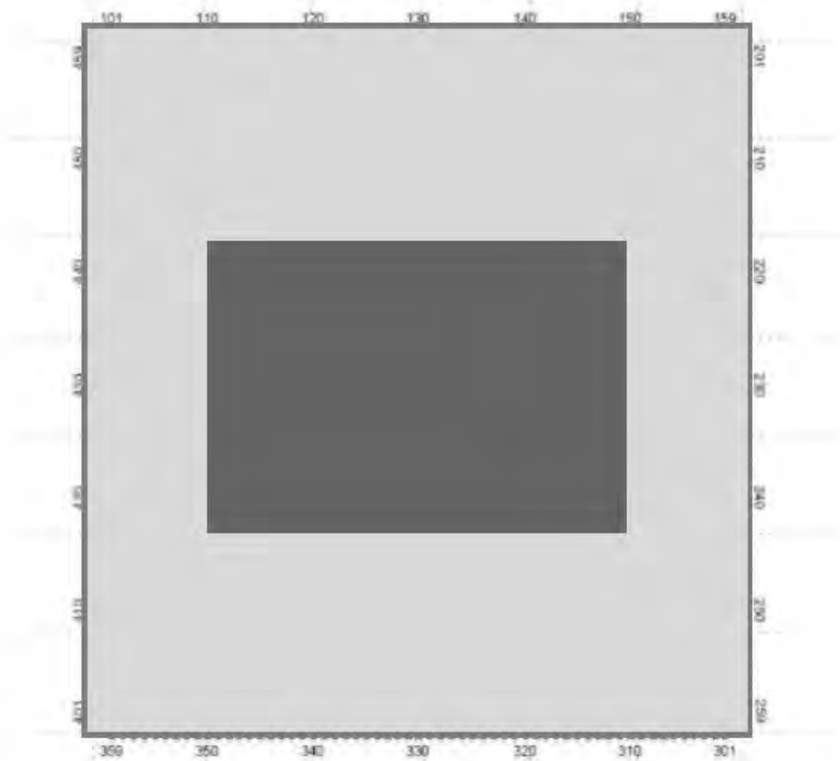


BUILDING ORGANIZATION





BUILDING ORGANIZATION



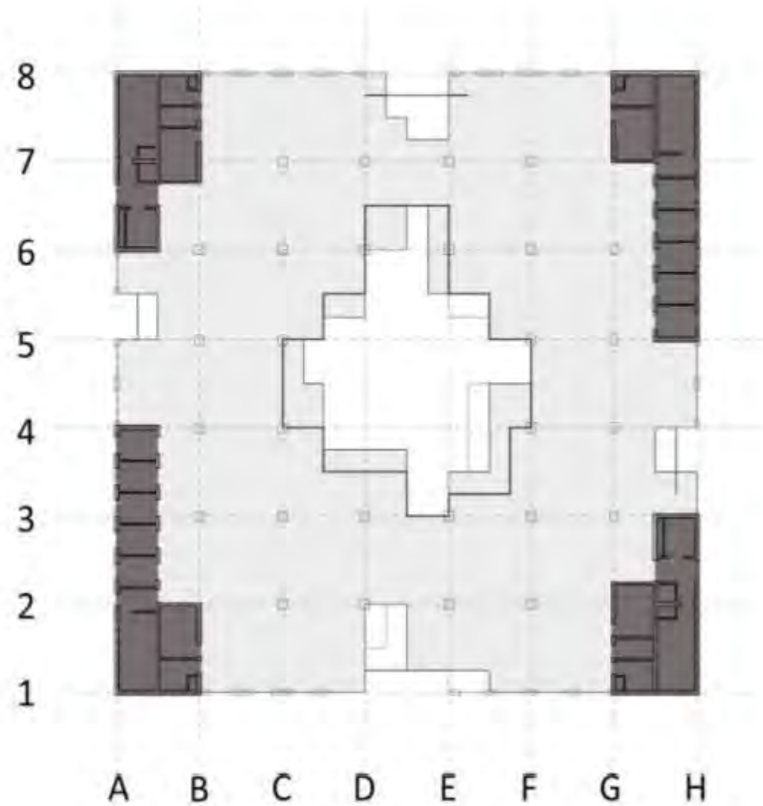
GENERIC PLAN

floorplan WTC

WTC, New York City,
Typical Central Core Highrise

only horizontal communication
Core as obstacle

strictly divided by floorplates



DEEP PLAN

floorplan KACST

KACST HQ, Riyadh,
Deep Plan

Atrium enables directors to see that everyone does their work

Accessibility of directors

communication inside; views to outside

KEY SPATIAL CONSTRAINTS

FAQs, Reference parameters to support early design understanding

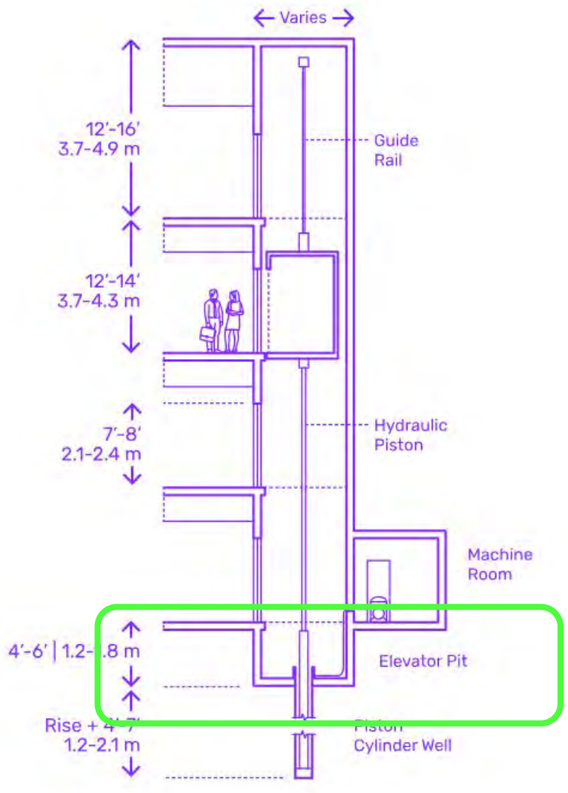
WIP

KEY SPATIAL CONSTRIANTS
Elevator

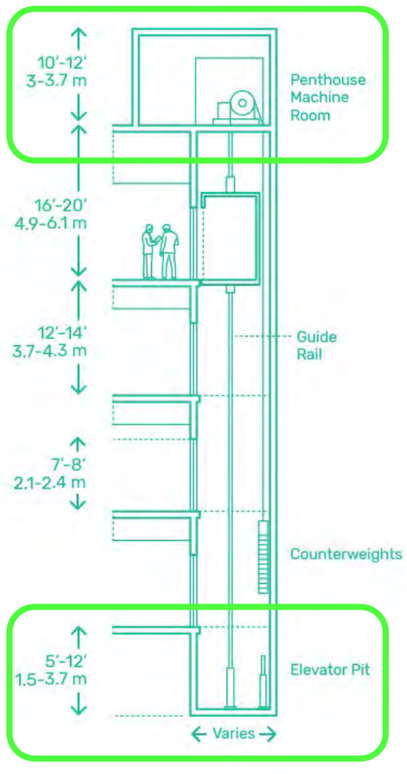
AQUELLUM - HOISTWAY INFORMATION																
								CAB		HOISTWAY						
S.No	Description	Lift Number	Lift Type	Configuration	No.of Lifts / Capacity (Kg)	Speed (m/s)	Approx Travel (m)	Clear Cab Sizes (w x d x h) (mm)	Clear Internal Entrance Sizes (w x h) (mm)	Structural Entrance Opening Sizes (w x h) (mm)	Clear Internal Hoistway Sizes (w x d) (mm)	Clear Pit Depth (mm)	Clear Overhead Height (mm)	Clear Machine Room Height (mm)	Occupied Space Below Lifts	Remarks
8	Duplex Residential	NE_RE12-RE13	Machine Room-Less	Duplex	1600	1.0	28	2100 x 1600 x 2700	1100 x 2400 SSCO - Front/rRear	1400 x 2550	**6150 x 2200**	1900	5000	N/A	Yes	

Considerations of Pit depth and overrun

- Affect on the space below and above the elevator



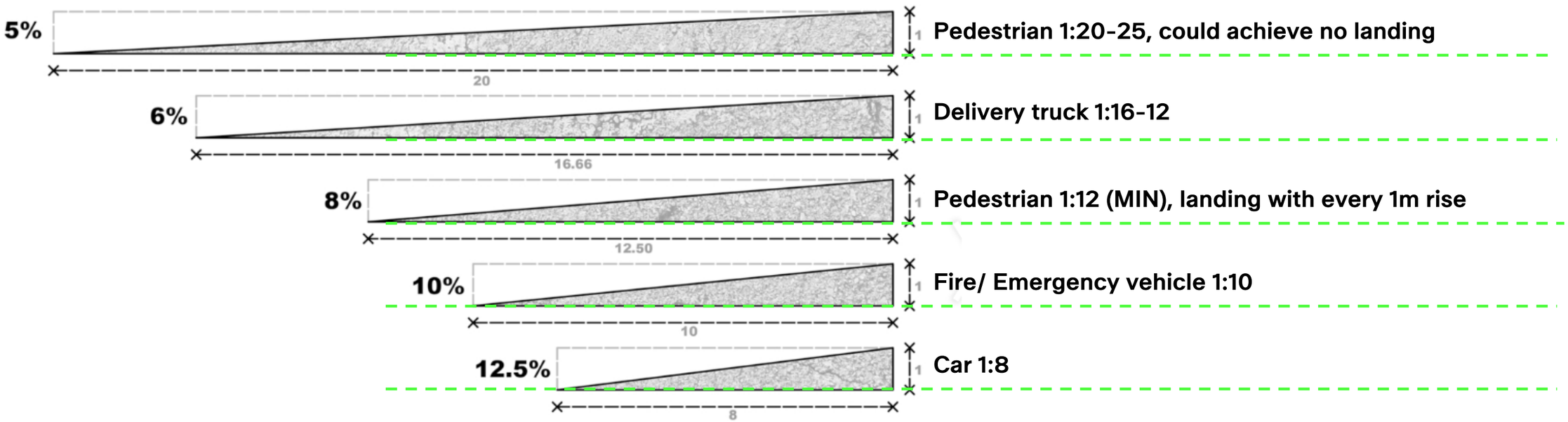
Hydraulic elevator



Traction elevator

***dimensions varies in types/ model of elevators*

KEY SPATIAL CONSTRIANTS
Ramp

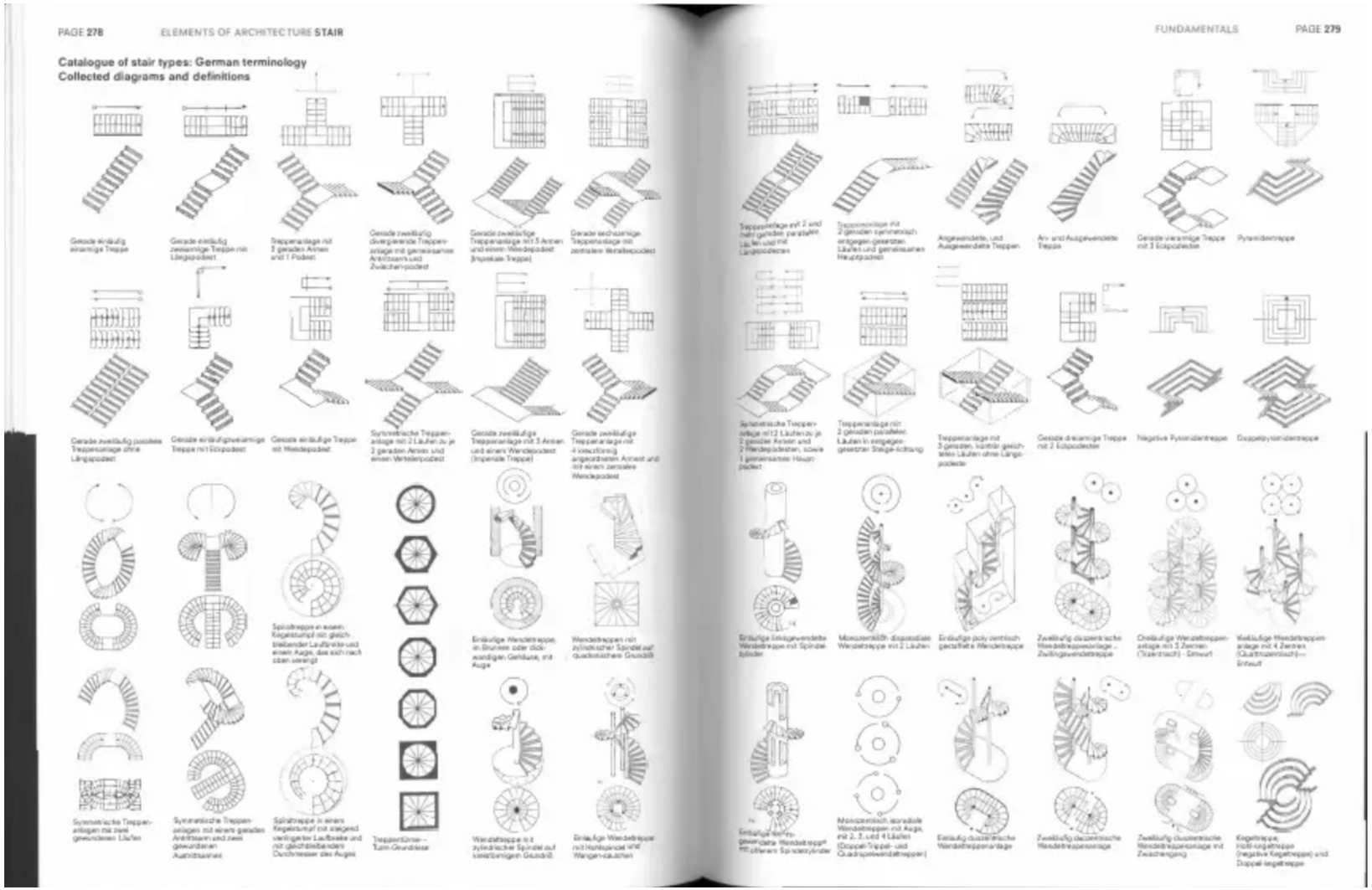


Ramp inclination
For initial length estimation to height

***dimensions varies in different building codes*

KEY SPATIAL CONSTRIANTS

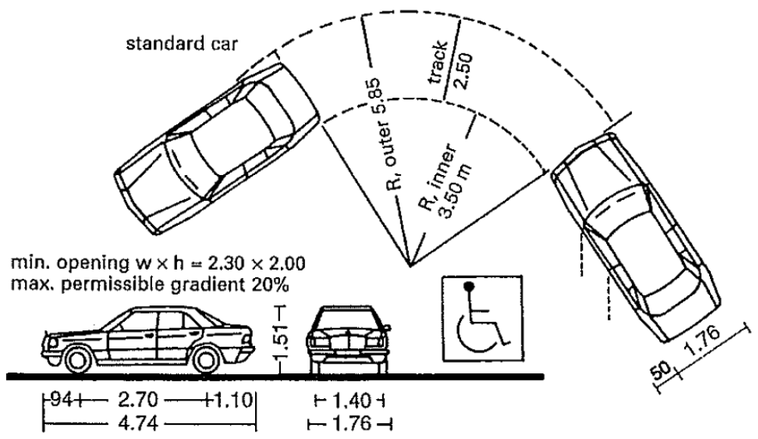
Stairs



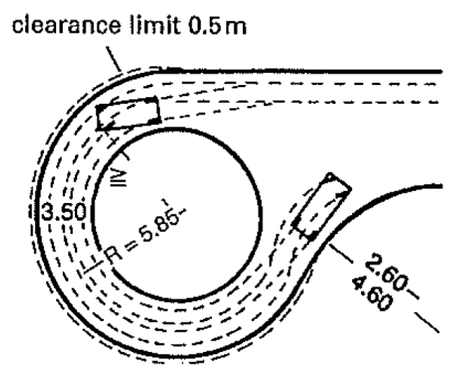
Stair Type	Typical Use	Riser (mm)	Tread / Going (mm)	Max Risers per Flight	Min Clear Width (mm)
Public Primary Stair	Main public circulation	**150–160**	**300–320**	16–18	**≥ 1800**
Public Secondary Stair	Normal public circulation	**160–170**	**280–300**	≤ 18	**≥ 1400**
Escape Stair (Fire)	Required egress	**180**	**280–300**	**≤ 16**	**≥ 1200, varies in programs, occupant load**
Spiral Stair (Non-Escape)	Secondary / limited public use	**170–200**	**≥ 230** *(measured at walking line)*	≤ 18	**≥ 800–1000**

L/\\//\

KEY SPATIAL CONSTRIANTS
Turning Radius



1 Standard car

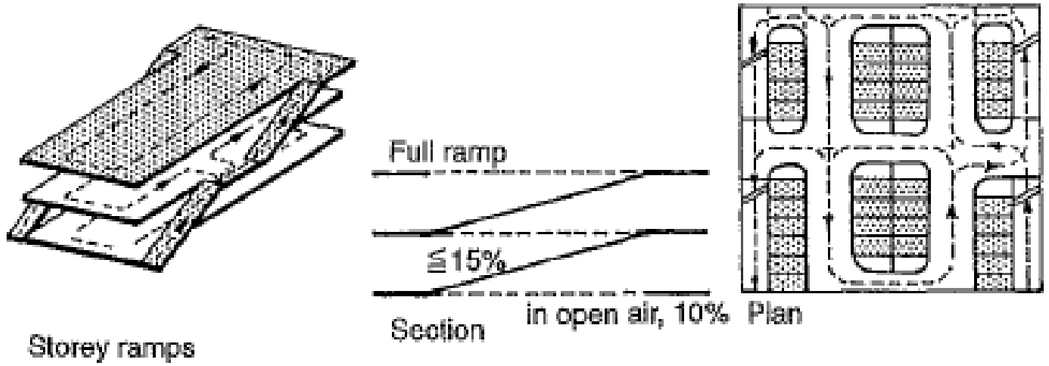


2 Turning circle of a car

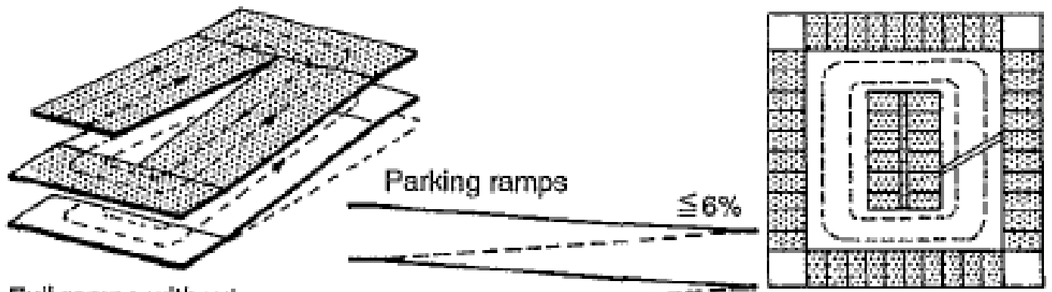
Vehicle Type	Inner Turning Radius (R_i)	Outer Swept Radius (R_o)	Notes / Early-Design Implication
Passenger Car	**3.5 m**	**≈ 5.8–6.0 m**	Neufert reference; suitable for parking ramps, drop-off loops
Van / Small Delivery	5.0–5.5 m	7.0–7.5 m	Typical service access; often underestimated
Transit Bus (12 m)	7.5–8.0 m	10.5–11.0 m	Bus bays, terminals, stations
Coach / Long Bus (15 m)	9.0–9.5 m	12.5–13.5 m	Intercity / tour bus circulation
Fire Engine (Large)	10.0–10.5 m	14.0–15.0 m	Critical life-safety constraint — governs access roads
Rigid Truck (16–18 m)	11.0–11.5 m	15.0–16.0 m	Loading docks, service yards
Articulated Truck / Trailer	13.0–13.5 m	17.0–18.0 m	Logistics / industrial access

***dimensions might change depends on requirement*

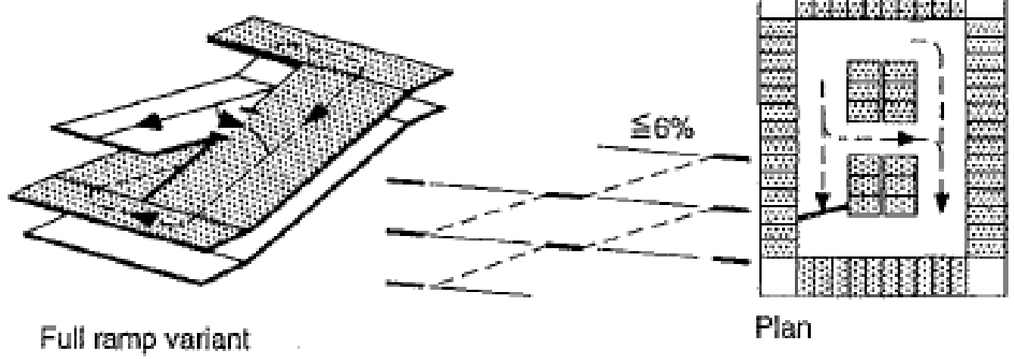
KEY SPATIAL CONSTRIANTS
Parking examples



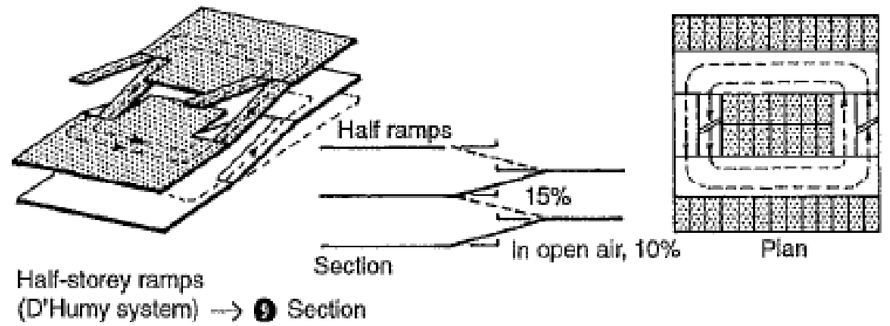
Storey ramps



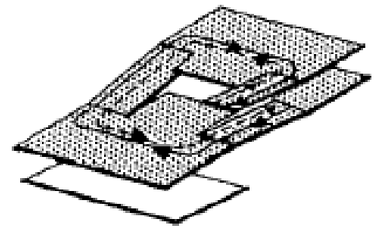
Full ramps without loss of space, gradient ≦6%



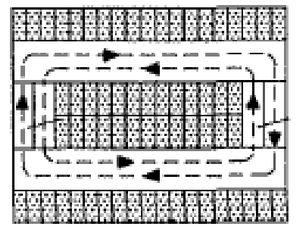
Full ramp variant

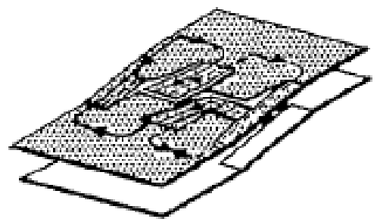


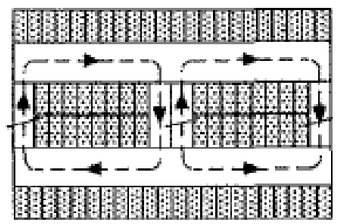
Half-storey ramps (D'Humy system) → ❾ Section

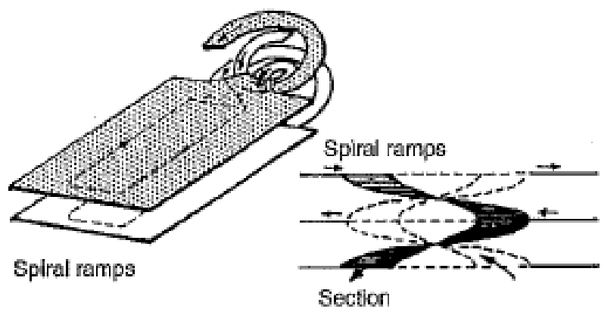


D'Humy ramp variants





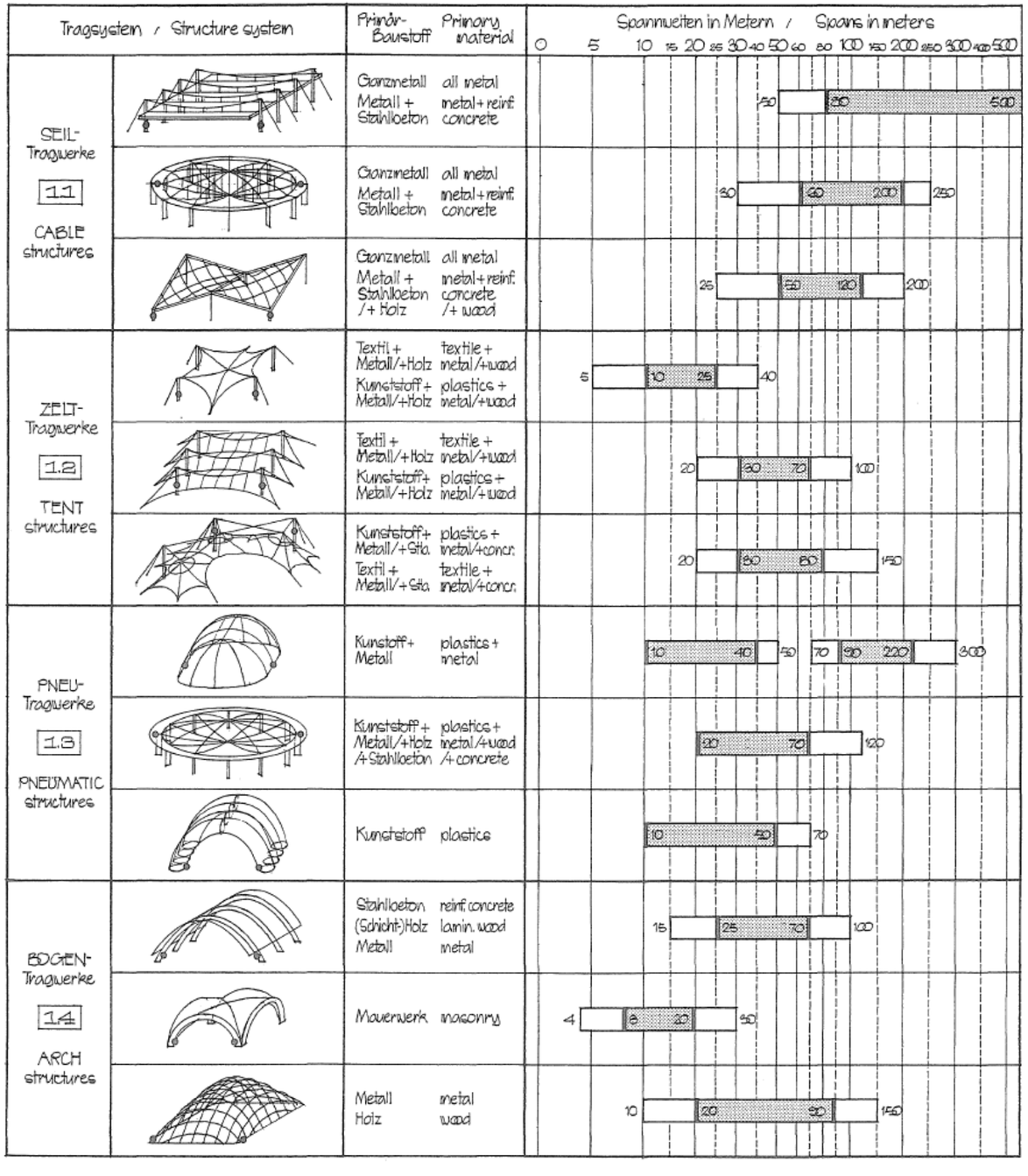


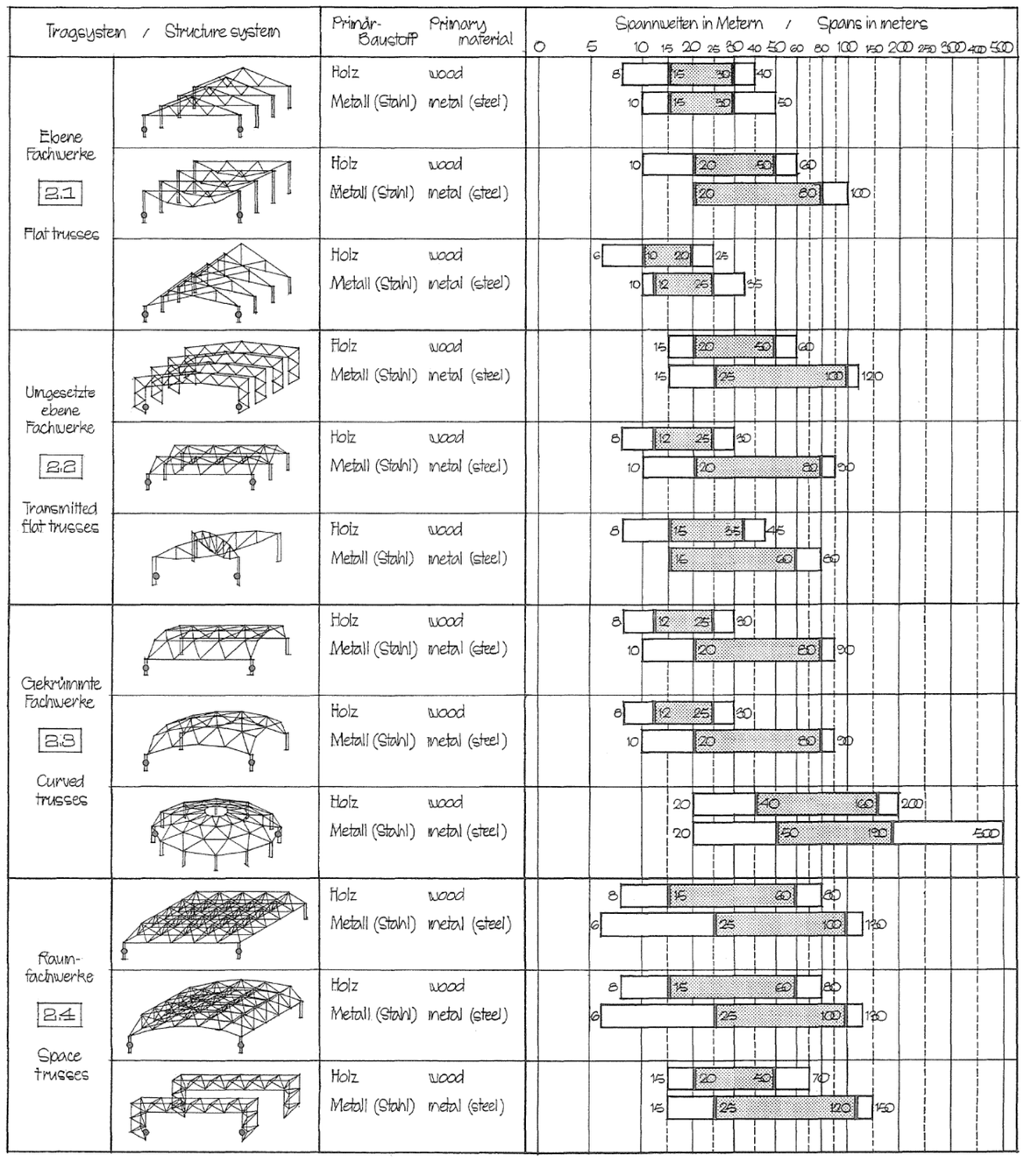


Spiral ramps

KEY SPATIAL CONSTRIANTS

Structural span

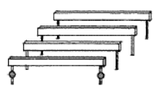
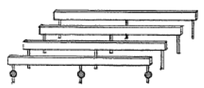
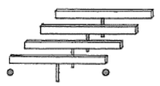
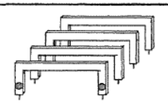
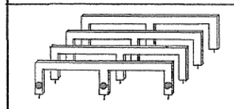
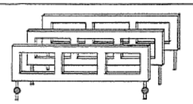
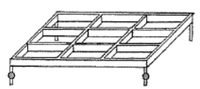
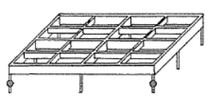
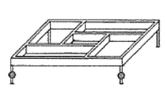
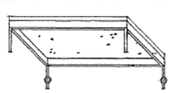
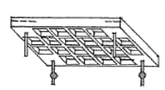
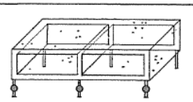


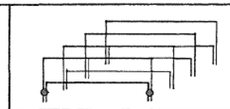
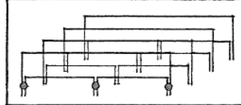
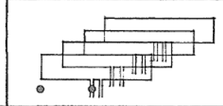
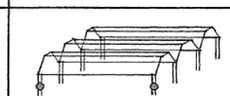
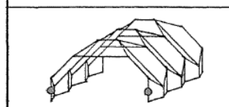
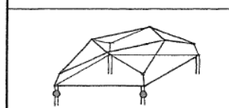
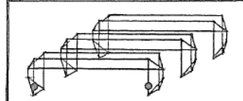
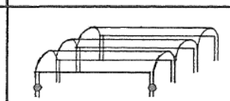
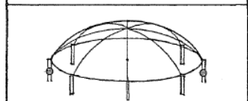
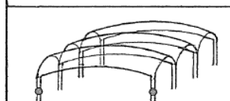
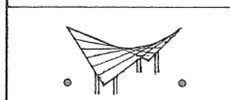
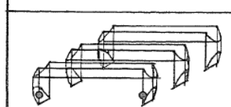


LAVA

KEY SPATIAL CONSTRIANTS

Structural span

Tragsystem / Structure system		Primär-Baustoff	Primary material	Spannweiten in Metern / Spans in meters
BALKEN-Tragwerke 3.1 BEAM structures		Holz	wood	4 8 12
		Metall (Stahl)	metal (steel)	5 7 20 25
		Stahlbeton	reinf. concrete	4 10 15
		Leimholz	glued wood	7 10 30 35
		Metall (Stahl)	metal (steel)	5 8 25 30
		Spannbeton	stressed concr.	7 10 25 30
		Holz	wood	4 8 12
		Metall (Stahl)	metal (steel)	5 7 20 25
		Stahlbeton	reinf. concrete	4 8 12
RAHMEN-Tragwerke 3.2 FRAME structures		Leimholz	glued wood	10 15 40 50
		Metall (Stahl)	metal (steel)	10 15 60 80
		Stahlbeton	reinf. concrete	7 10 25 30
		Leimholz	glued wood	10 15 45 55
		Metall (Stahl)	metal (steel)	10 15 65 75
		Stahlbeton	reinf. concrete	8 10 28 35
		Leimholz	glued wood	15 20 50 60
		Metall (Stahl)	metal (steel)	15 20 70 90
		Stahlbeton	reinf. concrete	10 15 50 40
BALKENROST-Tragwerke 3.3 BEAM GRID structures		Leimholz	glued wood	10 12 25 30
		Metall (Stahl)	metal (steel)	10 12 25 30
		Stahlbeton	reinf. concrete	5 8 16 20
		Leimholz	glued wood	10 15 30 35
		Metall (Stahl)	metal (steel)	10 15 30 35
		Stahlbeton	reinf. concrete	5 8 20 25
		Leimholz	glued wood	8 10 20 25
		Stahlbeton	reinf. concrete	5 8 15 18
PLATTEN-Tragwerke 3.4 SLAB structures		Holz (-bohlen)	wood (planks)	0 3 6
		Stahlbeton	reinf. concrete	0 6 8
		Stahlbeton	reinf. concrete	5 7 15 20
		Stahlbeton	reinf. concrete	5 4 8 12

Tragsystem / Structure system		Primär-Baustoff	Primary material	Spannweiten in Metern / Spans in meters
SCHEIBEN-Tragwerke 4.1 PLATE structures		Stahlbeton	reinf. concrete	8 10 40 50
		Holz	wood	6 15 30 50
		Stahlbeton	reinf. concrete	10 15 30 60
		Holz	wood	8 10 40 50
		Stahlbeton	reinf. concrete	5 8 20 25
		Holz	wood	3 5 15 20
FALTWERK Tragwerke 4.2 FOLDED PLATE structures		Stahlbeton	reinf. concrete	10 15 30 60
		Holz	wood	8 10 40 50
		Stahlbeton	reinf. concrete	20 25 150 200
		Holz	wood	15 20 120 150
		Stahlbeton	reinf. concrete	20 25 80 100
		Holz	wood	15 20 60 80
		Stahlbeton	reinf. concrete	10 20 70 90
		Holz	wood	10 15 60 70
SCHALEN-Tragwerke 4.3 SHELL structures		Stahlbeton	reinf. concrete	10 20 60 75
		Stahlbeton	reinf. concrete	20 40 150 200
		Stahlbeton	reinf. concrete	15 25 70 90
		Stahlbeton	reinf. concrete	15 25 60 70
		Holz	wood	15 20 50 60
		Stahlbeton	reinf. concrete	20 25 80 100

LAVA

KEY SPATIAL CONSTRIANTS

Cores

Impact on façade articulation, circulation, planning efficiency, daylight

SIDE CORE

Advantages
Maximizes **uninterrupted floor area**
Enables **shallow plan depths**
Improves **daylight and facade access**

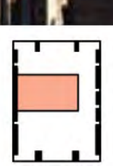
Disadvantages
Uneven **escape distances**
Structural asymmetry
Less redundancy in vertical circulation
Fire egress often requires **compensation strategies**

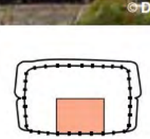
Typical Spatial Consequences
Linear circulation
Strong facade orientation
Clear zoning gradient

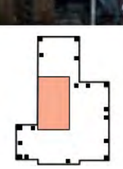






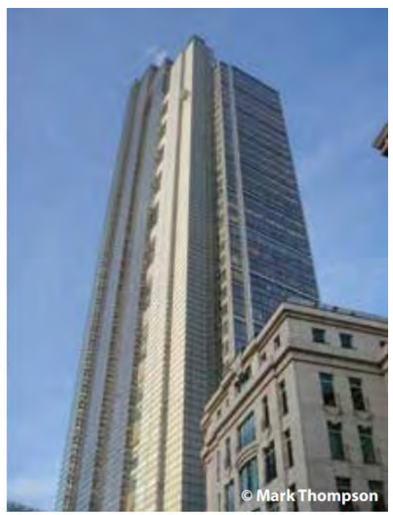


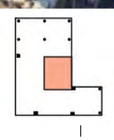




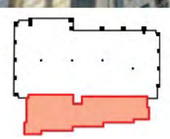












KEY SPATIAL CONSTRIANTS
Cores

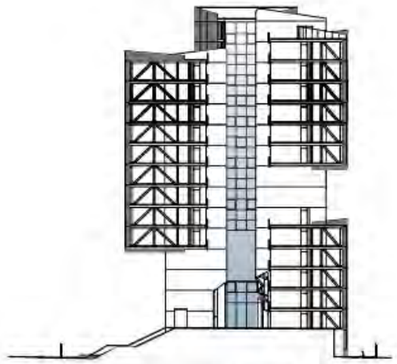
Impact on façade articulation, circulation, planning efficiency, daylight

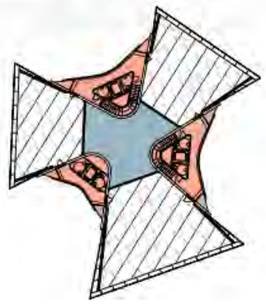
SPLIT CORE

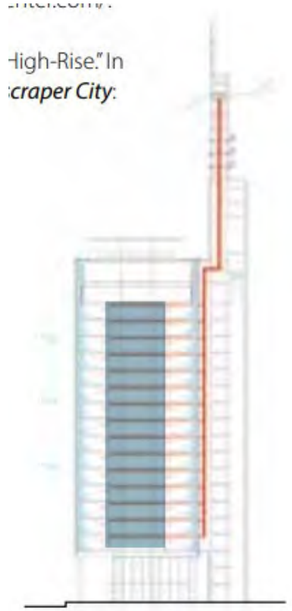
Advantages
Shorter **escape travel distances**
High **egress redundancy**
Strong **program separation** potential
Works well for **large or deep floorplates**

Disadvantages
Increased **area loss**
More complex **structural and MEP coordination**
Higher construction and operational cost

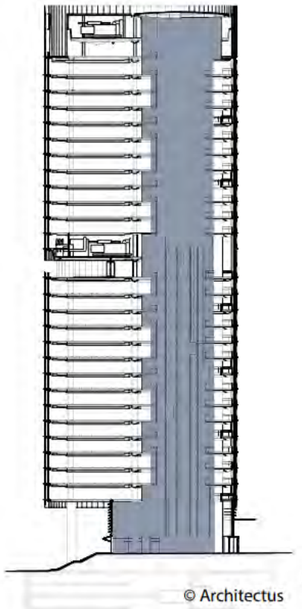
Typical Spatial Consequences
Zonal planning
Multiple circulation anchors
Flexible compartment layouts

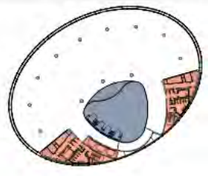

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KEY SPATIAL CONSTRIANTS
Cores

Impact on façade articulation, circulation, planning efficiency, daylight

CENTRAL CORE

Advantages
- Efficient **structural and service stacking**
- Clear **fire compartmentation**
- Balanced **escape distances** in all directions
- Simplifies **MEP riser coordination**

Disadvantages
- Reduces **continuous floor area**
- Can interrupt **large-span spaces**
- Daylight penetration may be reduced
- Less suitable for **linear or narrow plans**

Typical Spatial Consequences
- Ring-like circulation
- Even planning zones



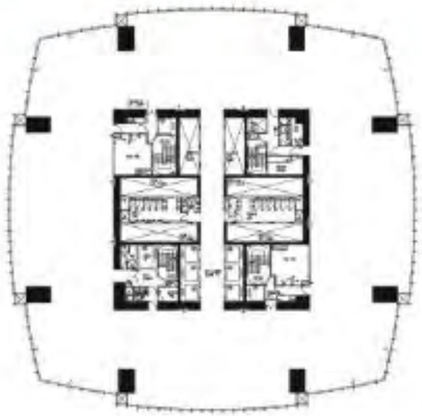
Taipei 101 Tower



Petronas Towers 1 and 2



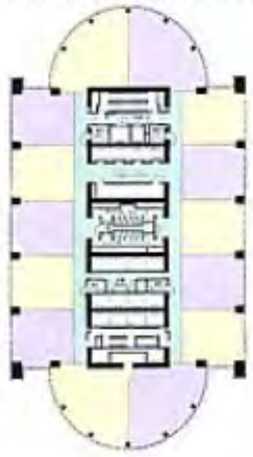
Jin Mao Tower



2 International Finance Centre

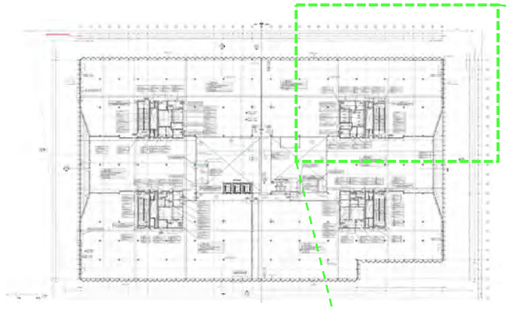


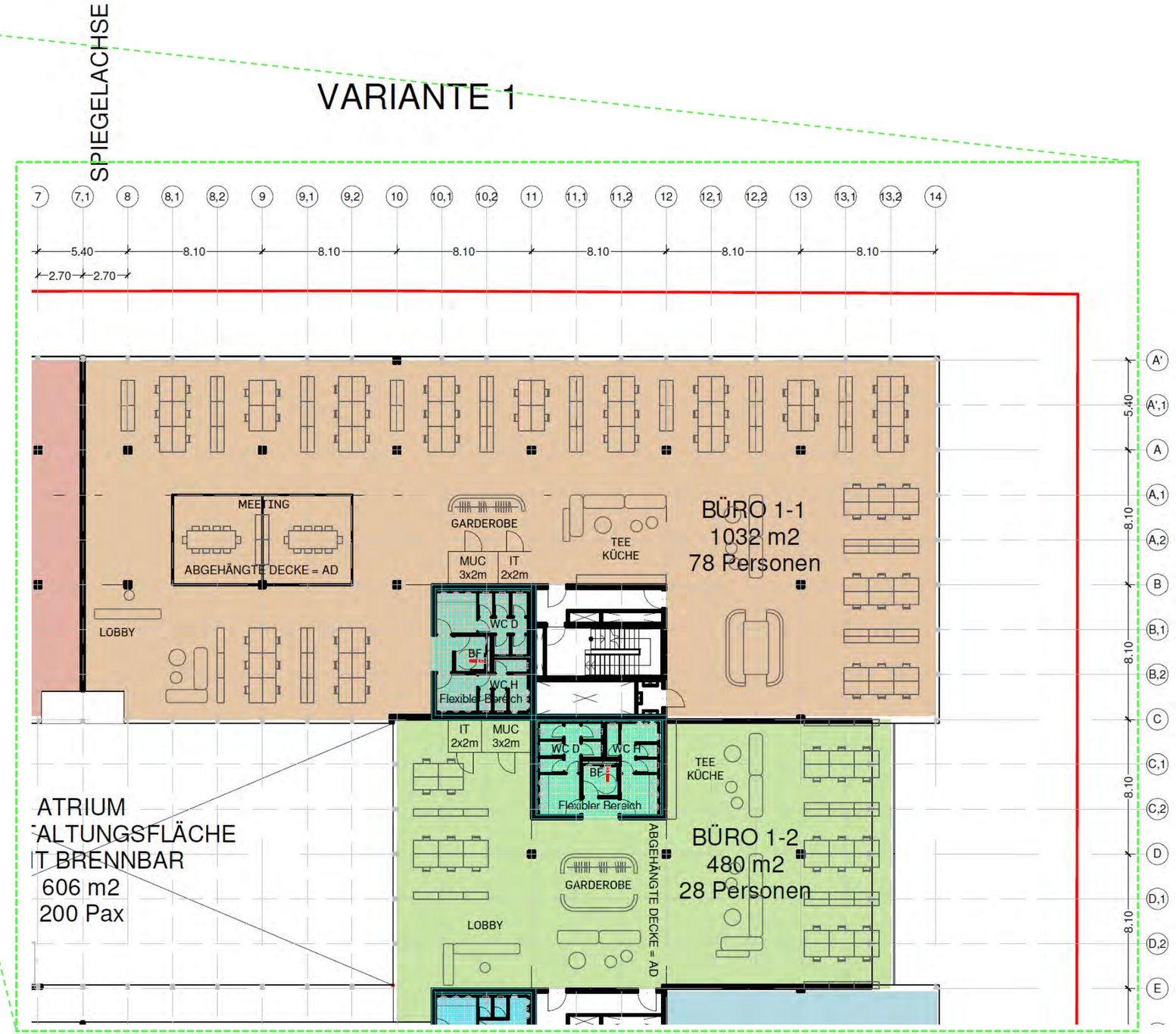
CITIC Plaza



Shun Hing Square

KEY SPATIAL CONSTRIANTS
Grid



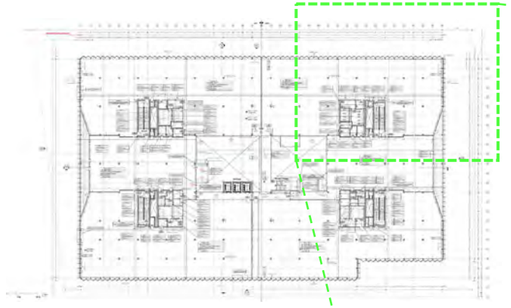


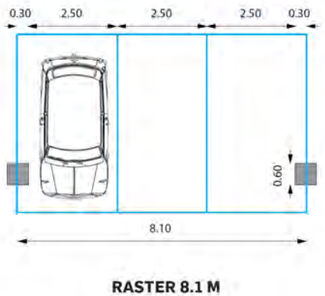
292 - Nalepaland, Berlin

Setting up grid

1) Consider the structural span
2) Influence the division of façade module
3) Influence the alignment of furniture layout, i.e. views and circulation
4) Parking

KEY SPATIAL CONSTRIANTS
Grid



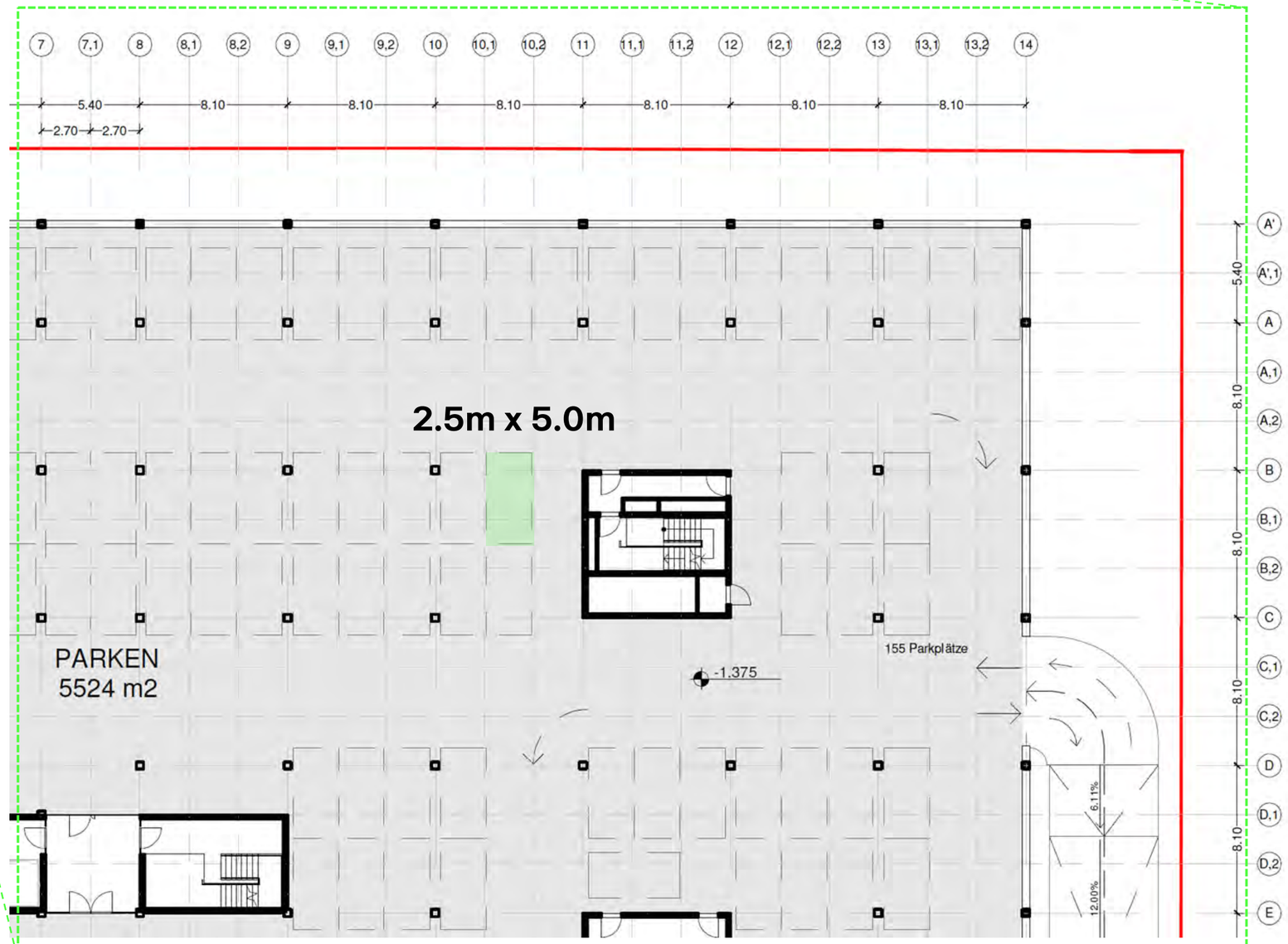


RASTER 8.1 M

292 - Nalepaland, Berlin

Setting up grid

1) Consider the structural span
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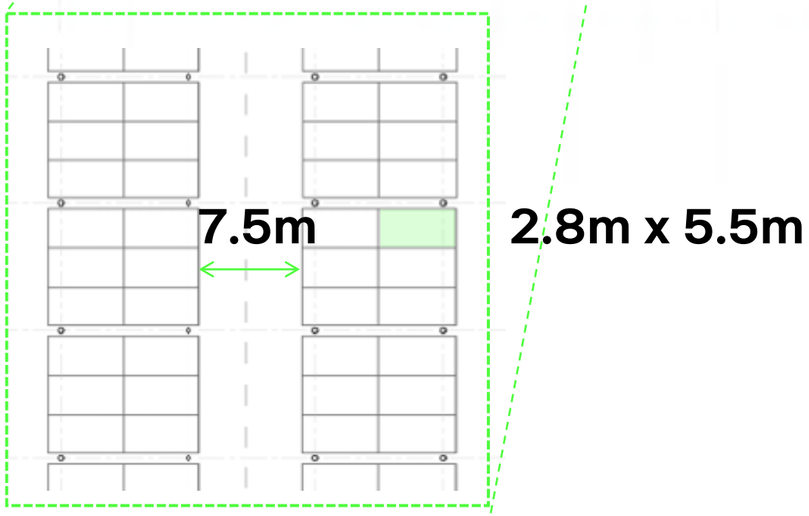


L/\V/\

**Parking requirements are varies from regions*

KEY SPATIAL CONSTRIANTS
Grid





497 - Hittin, KSA

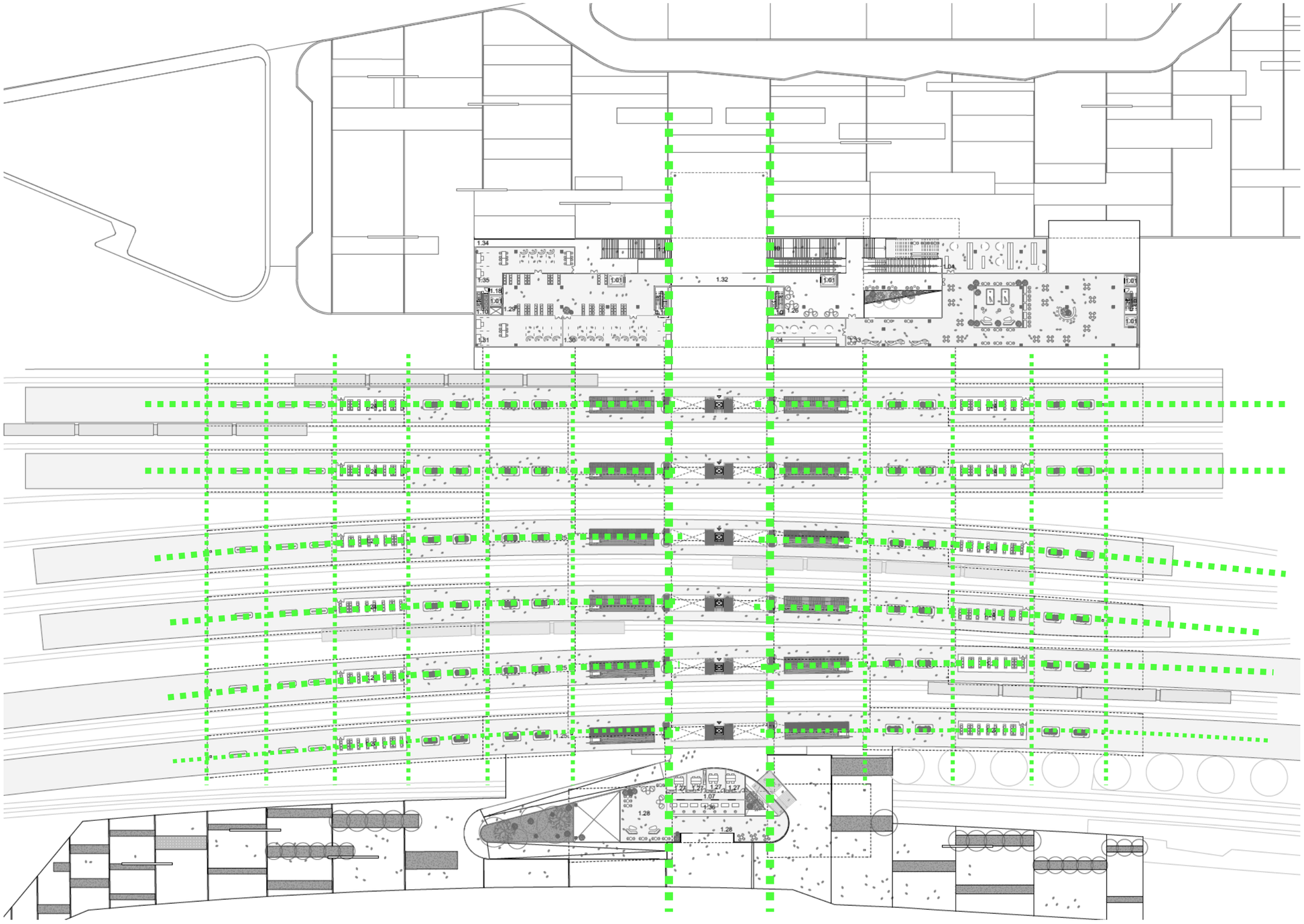
Setting up grid

1) Consider the structural span
2) Influence the division of façade module
3) Influence the alignment of furniture layout, i.e. views and circulation
4) Parking

L/\V/\

***Parking requirements are varies from regions*

KEY SPATIAL CONSTRIANTS
Grid

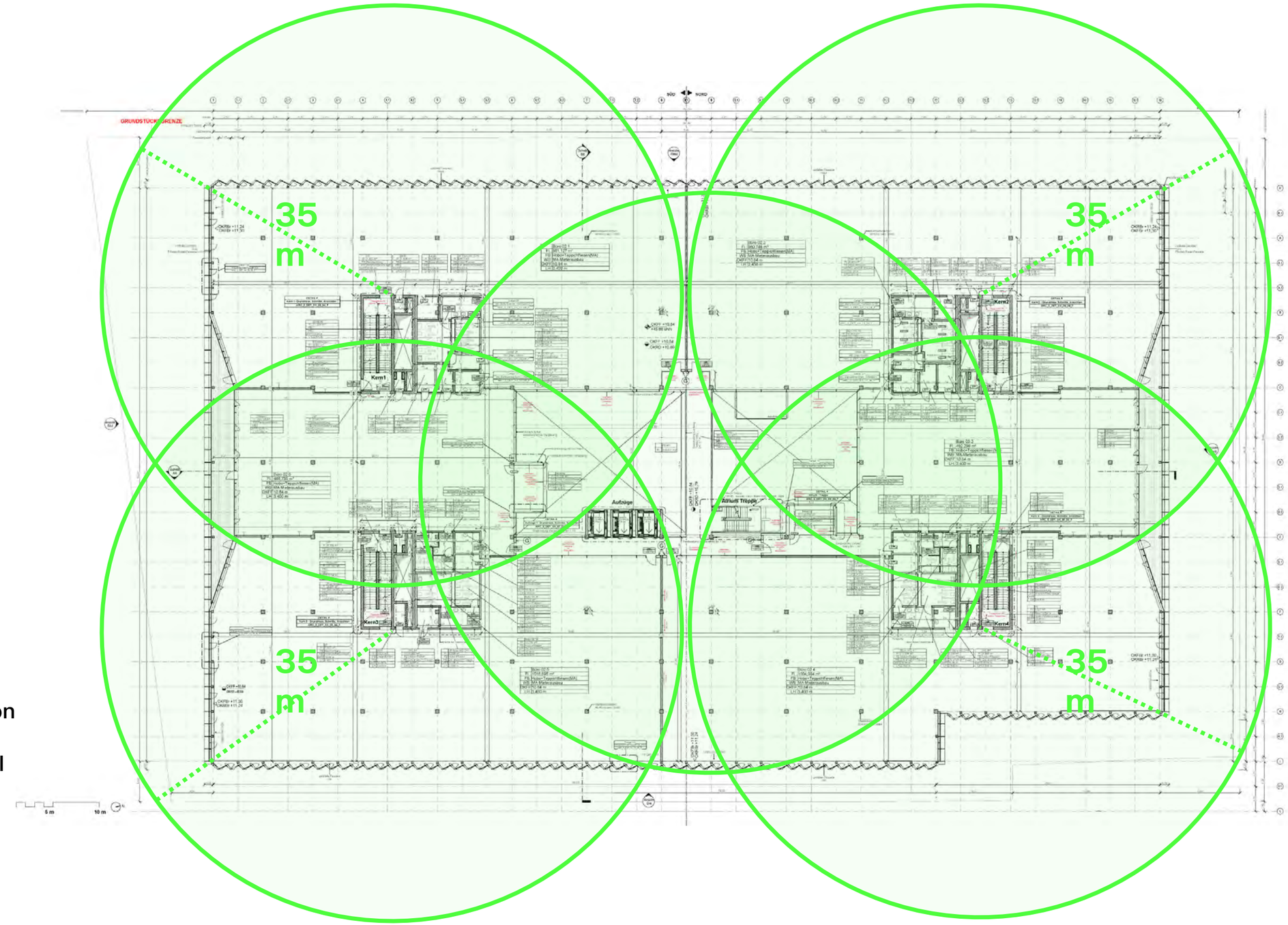


475 - ONCF, Morocco

Setting up grid

1) Structural grid in relation to the circulation flow
2) Structural grid inform the station’s canopy

KEY SPATIAL CONSTRIANTS
FLS



292 - Nalepaland, Berlin

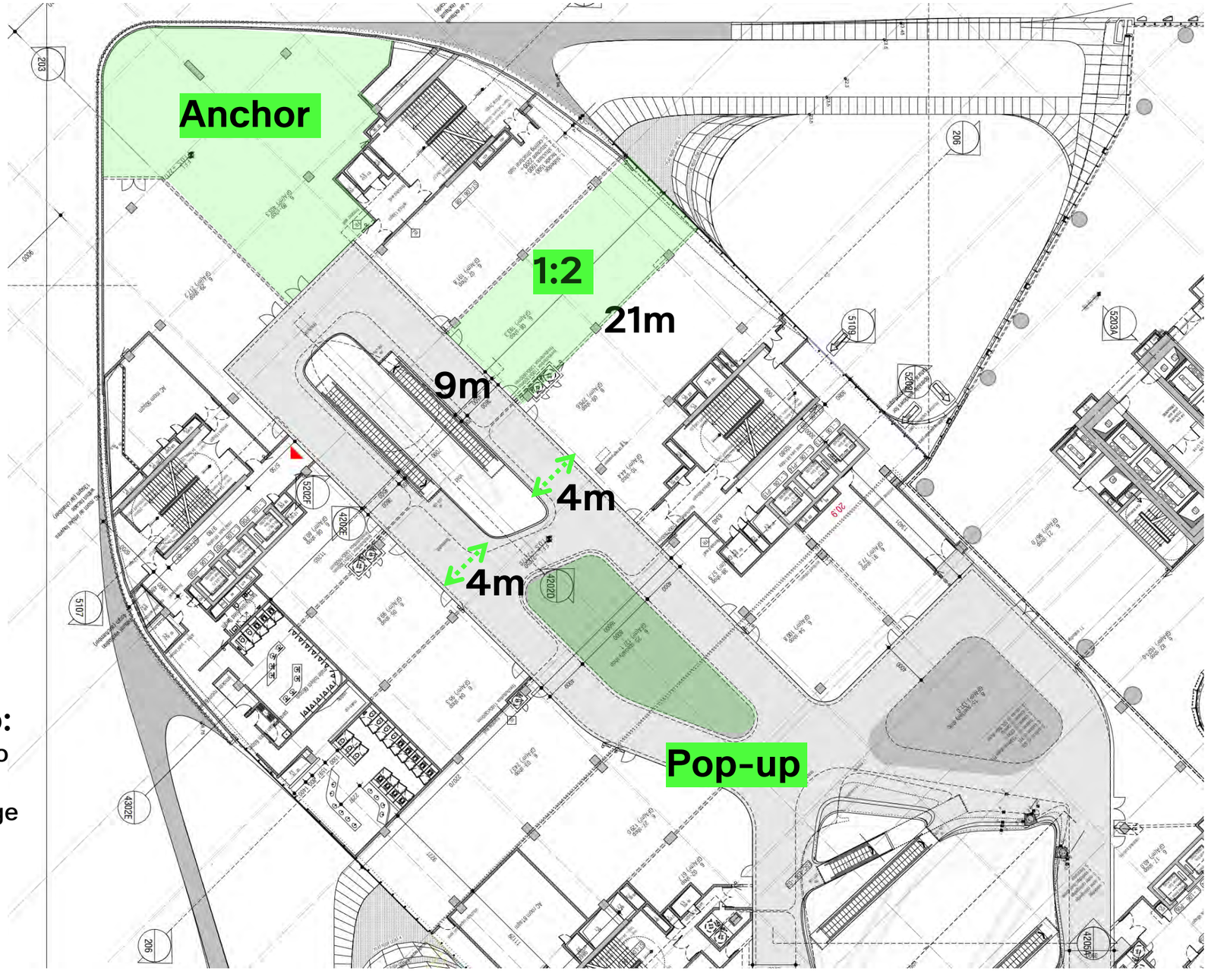
Setting up cores

1) Escape distances in relation to the depth of floor plan
2) Consideration as structural elements
3) Vertical circulation
4) Include risers for building services

L/\V/\

**Escape distance varies by building codes*

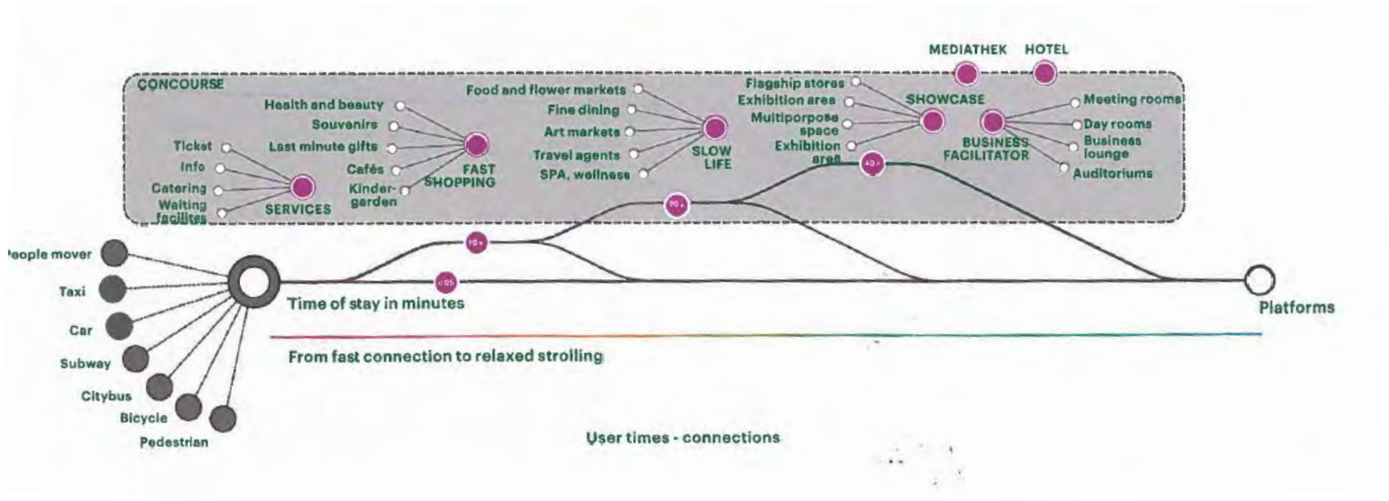
KEY SPATIAL CONSTRIANTS
Program proportions

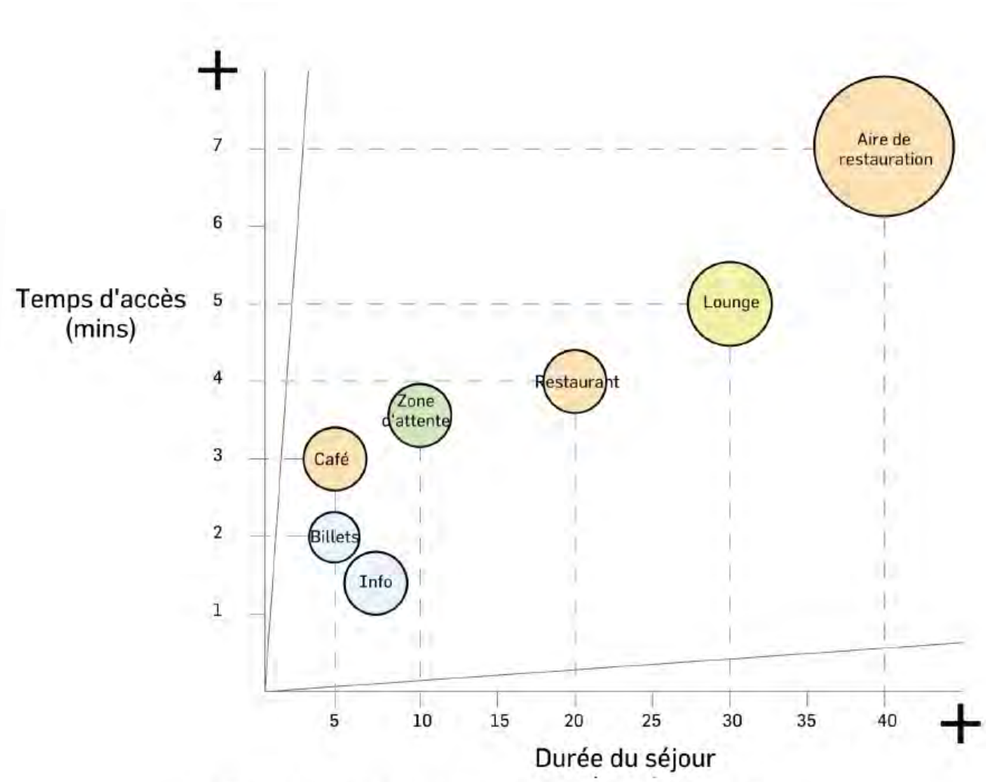


Shop front and depth ratio:

1) 1:2 to 1:3 as recommended ratio
2) Primary walkway width 4-6m
3) Anchor stores should encourage circulation
4) Pop up store should activate corridors

KEY SPATIAL CONSTRIANTS
Program distribution rule



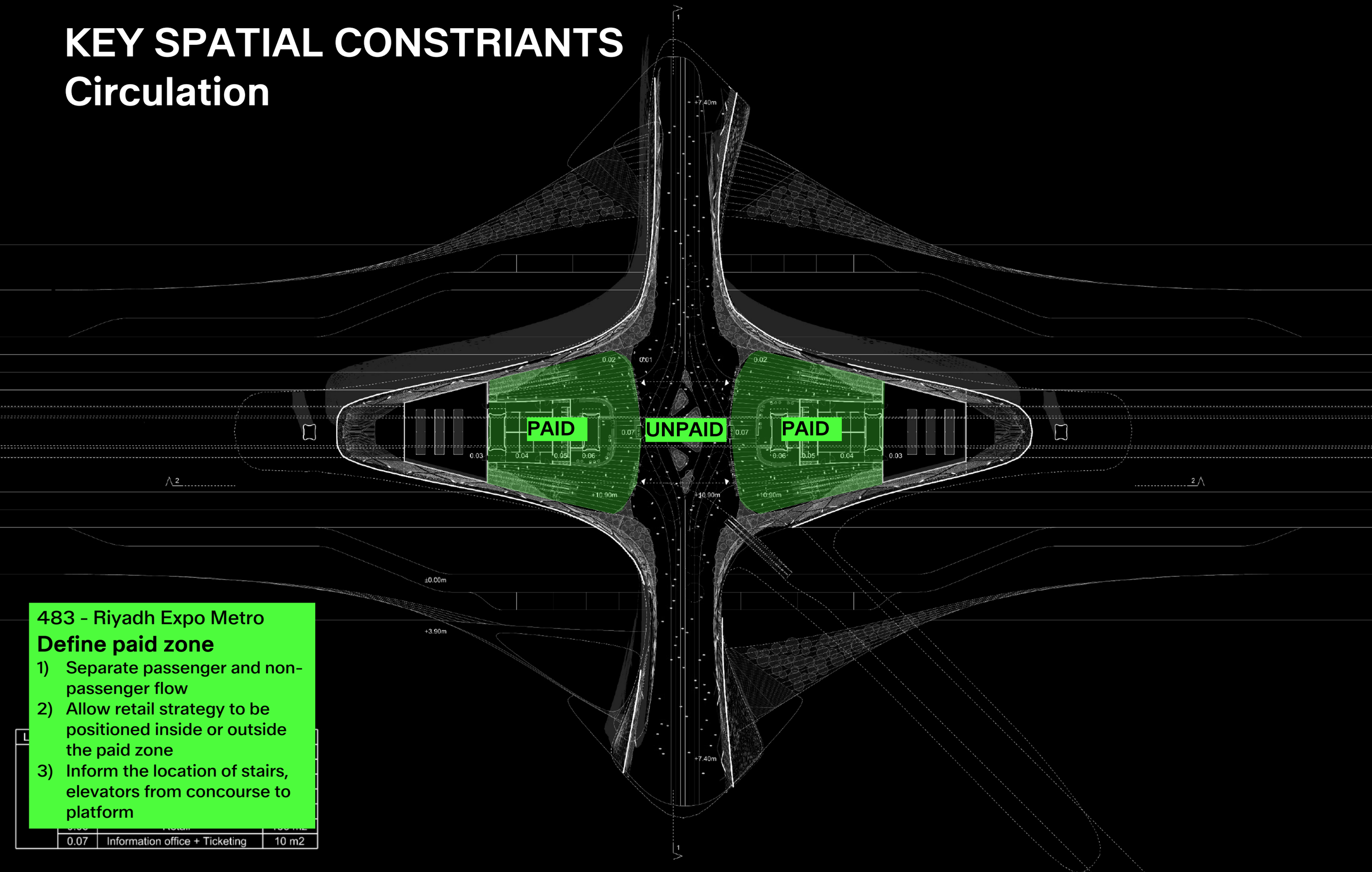


Répartition du programme basée sur le temps d'accès

Time based design

1) Identify typical passenger dwell times across station zones
2) Relate time spent to appropriate program types
3) Use time as parameter to guide where programs are placed along passenger flows

KEY SPATIAL CONSTRIANTS
Circulation



483 - Riyadh Expo Metro

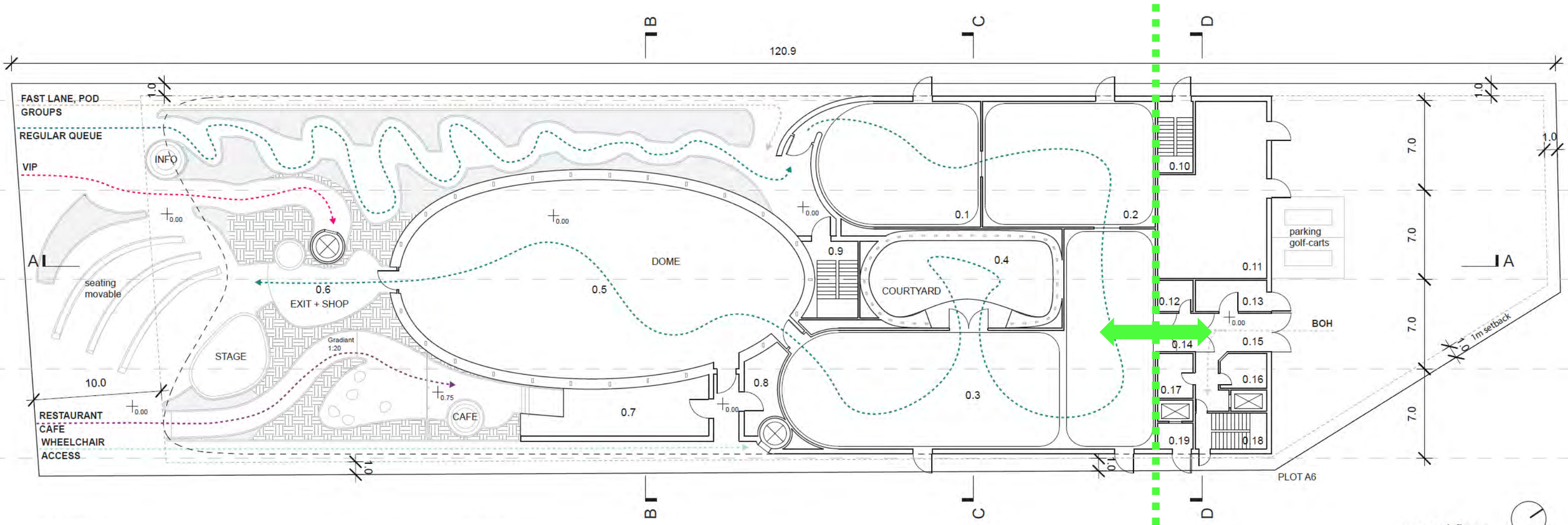
Define paid zone

1) Separate passenger and non-passenger flow
2) Allow retail strategy to be positioned inside or outside the paid zone
3) Inform the location of stairs, elevators from concourse to platform

0.07	Information office + Ticketing	10 m2

0 2 5 10m

KEY SPATIAL CONSTRIANTS
Circulation



428 – Kuwait pavilion, Osaka

Define FOH and BOH routing

1) Informed by site analysis, to define the site's public and non-public edge
2) The delivery route from BOH establish the interface from exhibition area to BOH
3) BOH routes do not intersect the public space

FOH 75%

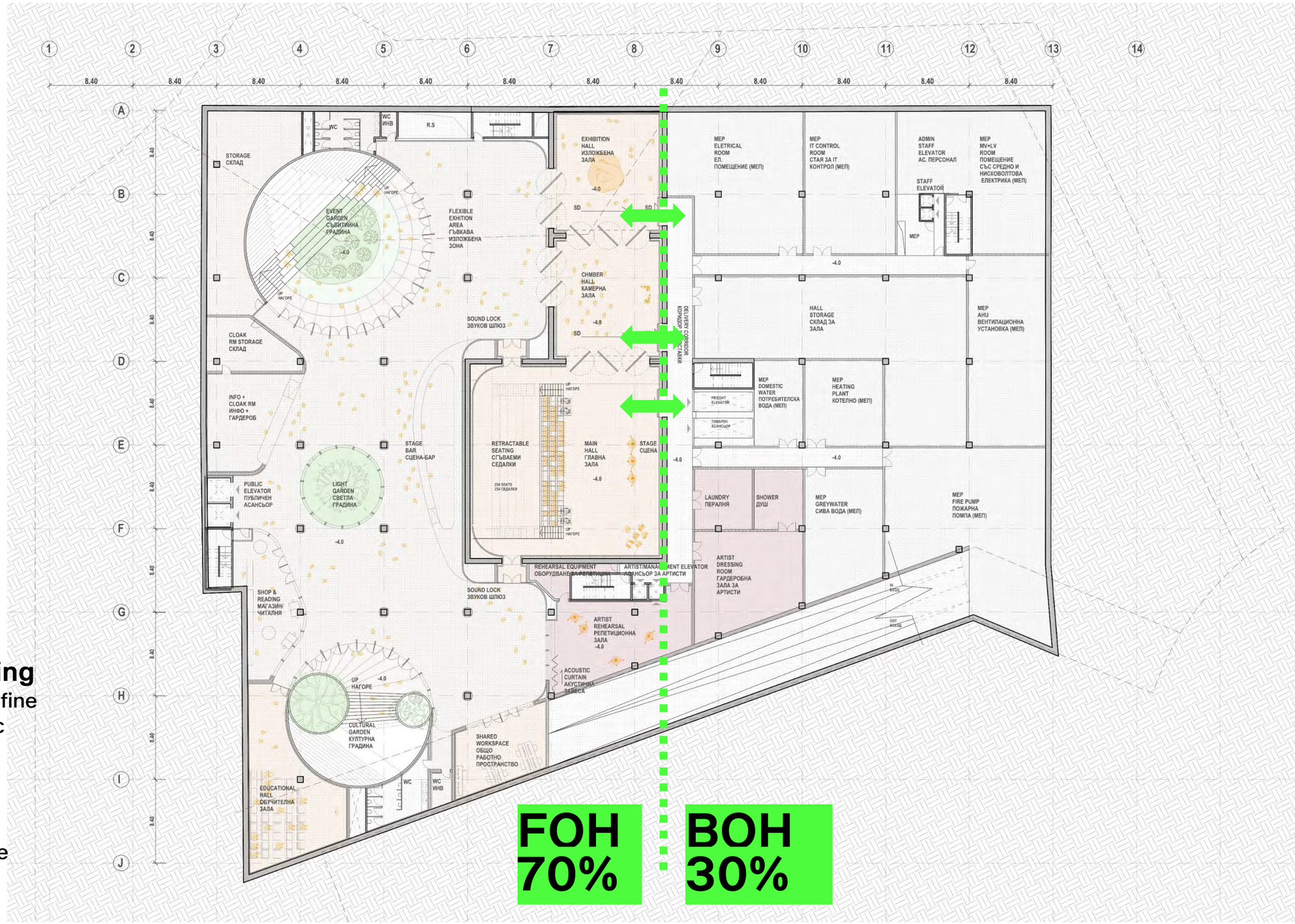
BOH 25%

KEY SPATIAL CONSTRIANTS
Circulation

499 – Bur, Burgas

Define FOH and BOH routing

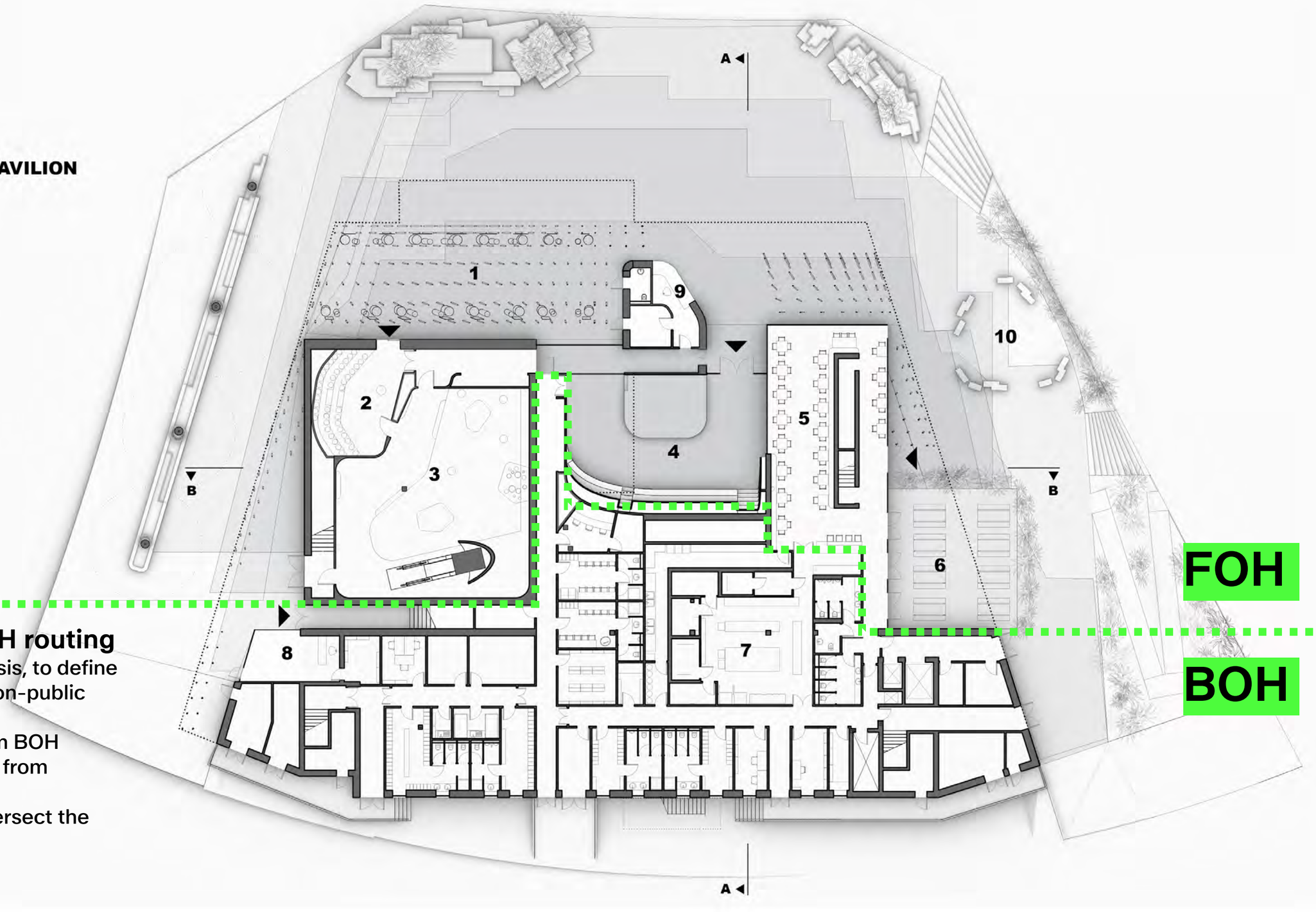
1) Informed by site analysis, to define the site's public and non-public edge
2) The delivery route from BOH establish the interface from exhibition area to BOH
3) BOH routes do not intersect the public space



KEY SPATIAL CONSTRIANTS
Circulation

EXPO 2020 GERMAN PAVILION
FLOOR PLAN
LEVEL 0

1 QUEUE
2 PRE SHOW
3 GERMANY BY NUMBERS
4 AUDITORIUM
5 RESTAURANT
6 EXTERIOR DINING
7 KITCHEN
8 GIFT SHOP
9 INFORMATION CENTER
10 EXTERIOR EVENT SPACE



284 – Expo, Dubai

Define FOH and BOH routing

1) Informed by site analysis, to define the site's public and non-public edge
2) The delivery route from BOH establish the interface from exhibition area to BOH
3) BOH routes do not intersect the public space

FOH

BOH